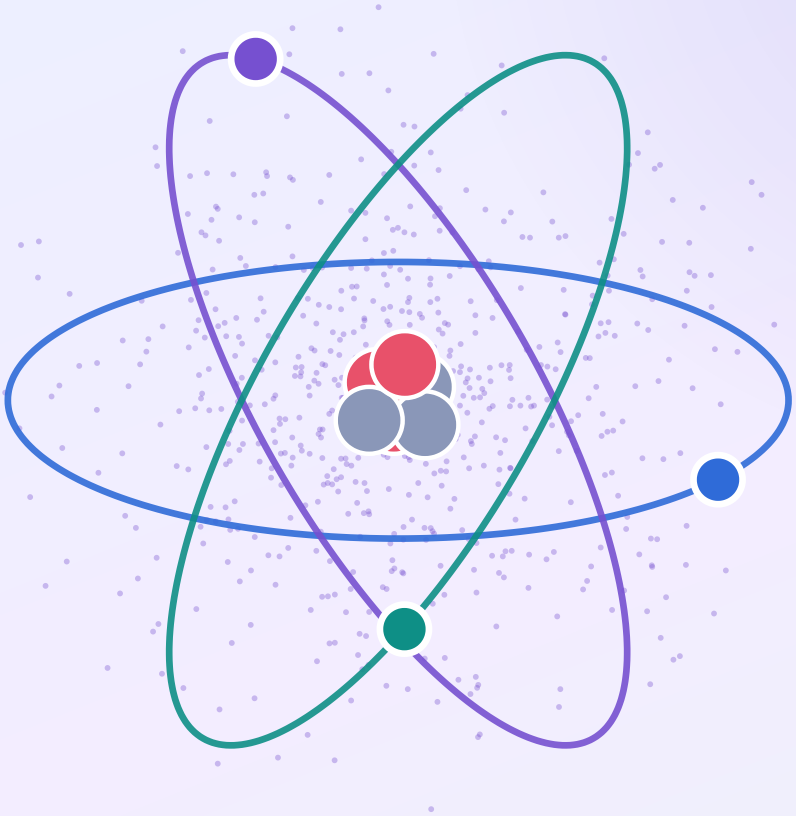


STRUCTURE OF ATOM

The complete handwritten notes, redesigned as an illustrated study book: theory, derivations, solved numericals, practice MCQs and NEET previous-year questions.

e^-	p^+	n^0	$h\nu$	λ	Δx	ψ
Electron	Proton	Neutron	Photon	Wavelength	Uncertainty	Wave function

About this chapter

Structure of Atom follows how the picture of the atom was built up: the discovery of the electron, proton and neutron; the atomic models of Thomson, Rutherford and Bohr; the wave and particle nature of radiation and matter; Heisenberg's uncertainty principle and Schrödinger's quantum mechanical model; and finally quantum numbers, the shapes of orbitals and the rules for filling them. The theory is followed by a large bank of solved exercise questions, three sets of practice MCQs and 65 NEET previous-year questions, all in the order of the original notes.

How the colours work

Definition
Key terms, stated exactly as in the notes

Concept
Important ideas, postulates and statements

Formula
Equations and relations on a grid-paper card

Observation
What experiments showed

Result
Conclusions and final answers

Remember
Points worth revising

Limitation
Failures and limitations of models

Numerical
Solved problems with every step of working

MCQ
Exam-style questions with all options

Faithful to the original notes

- This book reproduces all 40 pages of the handwritten notes. Every block carries a small marker such as Original notes p. N (N = page number) showing where it comes from.
- Values, steps and answers are reproduced exactly as written, even where they look inconsistent; nothing has been silently corrected.
- Words cut off at the edge of the scan and completed from context are shown with a dotted underline.
- Orange tags such as V1 mark readings to check against the original. Each one is explained in Appendix A.
- Where the notes leave an answer blank, the book says so instead of supplying one.
- Appendix B is the page-by-page verification checklist; Appendix C lists every diagram.

Contents

2.1 Atom and Sub-Atomic Particles 4

2.2 Discovery of Electron – Discharge Tube Experiment 4

2.3 Discovery of Proton 5

2.4 Discovery of Neutrons 6

2.5 Thomson's Model of Atom 6

2.6 Rutherford Model of Atom (1911) 7

2.7 Atomic Number and Mass Number 9

2.8 Isotopes, Isobars and Isotones 10

2.9 Wave Nature of Electromagnetic Radiations 10

2.10 Planck's Quantum Theory 13

2.11 Black Body Radiations 14

2.12 Photoelectric Effect 15

2.13 Atomic Spectra 17

2.14 Line Spectrum of Hydrogen 18

2.15 Bohr's Model of Atom 19

2.16 Dual Behaviour of Matter (De-Broglie's Equation) 22

2.17 Heisenberg Uncertainty Principle 24

2.18 Quantum Mechanical or Wave Mechanical Model of Atom 25

2.19 Quantum Numbers 26

2.20 Shapes of Atomic Orbitals 29

2.21 Filling of Orbitals in an Atom — Electronic Configuration 30

2.22 Solved Exercise Questions 33

2.23 Practice MCQs 55

2.24 Structure of Atom — NEET PYQs 61

A Appendix A — Items flagged for verification 77

B Appendix B — Page-by-page verification checklist 81

C Appendix C — Diagrams 84

Original notes p. 1

2.1 At The Atom

Atom and Sub-Atomic Particles

Atom

An atom is the smallest particle of an element that may or may not have independent existence. It is the basic building unit of matter.

Sub-Atomic Particles

An atom mainly consists of three particles:

Electron (e^-)
→ negatively charged

Proton (p^+)
→ positively charged

Neutron (n^0)
→ neutral

These particles together determine the structure and properties of an atom.

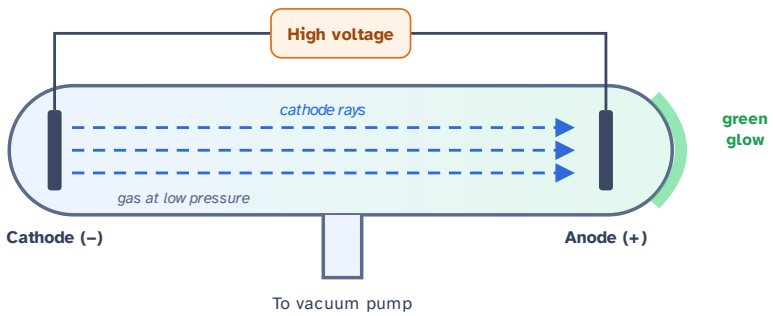
2.2 e Electron

Discovery of Electron – Discharge Tube Experiment

The electron was discovered through the discharge tube experiment.

Experimental setup

A cylindrical glass tube (50 cm long) V1, fitted with two metal electrodes i.e. cathode (–ve) and anode (+ve). The tube is filled with gas at low pressure and connected to a high voltage source and a vacuum pump to reduce pressure.



Discharge tube: cathode rays travel from cathode and make the opposite glass wall glow

Observations of the Experiment

1. At normal pressure (1 atm) gas does not conduct electricity even at high voltage i.e. no current flows.

   Conclusion: Gas is a poor conductor under normal conditions.

2. At low pressure, gas starts conducting electricity and a glow appears inside the tube. The colour of the glow depends on the nature of the gas.

3. At very low pressure, the glow disappears and the wall of tube opposite to the cathode starts glowing with a green light. This glow is called **fluorescence**.

Reason of fluorescence

This happens due to invisible rays coming from the cathode, which strike the glass wall. These rays are called **cathode rays**.

Properties of Cathode Rays

1 Move in straight lines	2 Cast shadow	3 Have mass and rotate paddle
4 Negatively charged	5 Produce heat	6 Ionise gases
7 Produce X-rays	8 Show fluorescence	9 Affect photographic plate

Conclusion of the experiment

From all these observations, it was concluded that

- Cathode rays are made up of material particles.
- These particles carry negative charge.
- These particles are called **electrons**.

Charge and Mass of Electron

Charge

Charge on electron $= 1.602 \times 10^{-19}$ C

Nature of charge = Negative

Mass of electron

Calculated using: e/m ratio (Thomson); charge (Millikan oil drop experiment)

Formula: Mass of electron $(m) = \dfrac{e}{e/m}$

Value of electron $= 9.1 \times 10^{-31}$ kg

2.3
p
Proton

Discovery of Proton

Atoms are electrically neutral, so presence of positive particles was expected.

Goldstein discovered protons in 1886 by using a modified discharge tube in which a perforated (holes) cathode was used.

At low pressure, new rays moving opposite to cathode rays were observed. These rays passed through holes of cathode and moved from anode to cathode direction and are called **Anode rays / Canal rays**.

Properties of Anode Rays

1. Travel in straight lines
2. Made of material particles (have mass)
3. Carry positive charge
4. Deflected by electric and magnetic fields but direction is opposite to cathode rays
5. Charge/mass depends upon nature of gas
6. These particles may carry +1, +2 or +3 charge

Original notes p. 2

2.4 n Neutron

Discovery of Neutrons

Neutrons were discovered by **James Chadwick in 1932**. He performed a scattering experiment. He bombarded light elements (Be, B) with α-particles.

Observation

The emitted particles have no charge (neutral) and their mass is equal to that of proton. These particles were called **neutrons**.

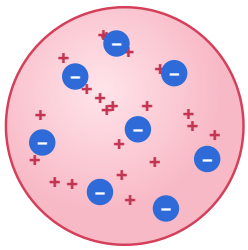
2.5 Th Thomson

Thomson's Model of Atom

In **1898 J.J. Thomson** proposed an atomic model.

According to this model

- An atom is a sphere of positive charge and electrons are embedded inside it like seeds in watermelon or plum pudding.
- The positive charge is uniformly spread.
- Electrons are fixed inside the sphere.
- Atom is overall electrically neutral.



Electrons embedded in a sphere of positive charge

Limitation

This model could not explain

(i) Stability of atom

(ii) Later experimental results.

Hence this model was rejected.

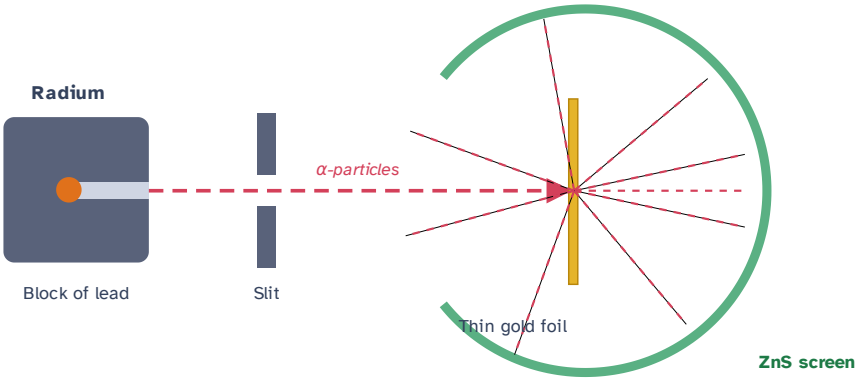
2.6
Ru
Rutherford

Rutherford Model of Atom (1911)

This model was proposed by **Ernest Rutherford in 1911** to explain structure of atom. It is based on the α-scattering experiment.

α-Scattering Experiment

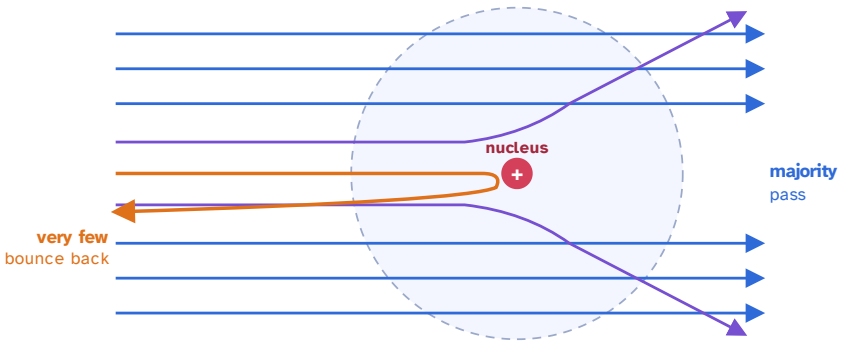
In this experiment, α-particles emitted from a radioactive source (radium) were passed through a lead slit and directed toward a thin gold foil which was surrounded by a zinc sulphide screen.



Rutherford's α-scattering set-up (radium in block of lead, slit, thin gold foil, ZnS screen)

Observations

1. Most of α-particles (99%) passed straight through gold foil and show no deflection.
2. Some α-particles deflected by small angles.
3. Very few (1 in 20000) were deflected back.



Path of α-particles near the nucleus: majority pass undeflected; very few bounce back

Conclusion

1. Most of the space inside an atom is empty because most particles passed without deflection.
2. Small deflections show the presence of positive charge.
3. Large deflections indicate a small, dense, positively charged centre.

Rutherford regarded this small, dense, heavy and positively charged centre inside an atom as the **nucleus**. All protons were supposed to be present in the nucleus. The space which surrounded the nucleus was called the **extra nuclear**

portion and all electrons were supposed to be present in it.

Features of Rutherford Model of Atom

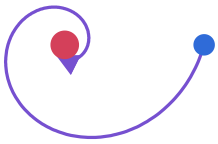
1. According to Rutherford Model, an atom consists of two parts: (a) Nucleus and (b) Extra nuclear part.
2. Nucleus is a heavy, small and +vely charged body present in the centre of an atom.
3. The positive charge of nucleus is due to protons.
4. Almost total mass of atom is concentrated in nucleus.
5. Electrons are present in extra nuclear portion.
6. Total +ve charge on protons is equal to total –ve charge on electrons.
7. The radius of atom is 10^{-10} m and nucleus is 10^{-15} m.
8. In an atom, the electron is under the influence of two types of forces: centripetal force (inward force) and centrifugal force (outward force).
9. The electrons in extra nuclear portion are not stationary but revolve around the nucleus at high speed in circular paths called **orbits**.

This model of atom is similar to the solar system, where the nucleus is like the sun and electrons may be compared to planets. The electrons are therefore called **planetary electrons**.

Original notes p. 3

Failure of Rutherford Model

1. **Stability Problem:** According to Rutherford model, electrons revolve around the nucleus. A revolving electron is an accelerating charged particle. As per electromagnetic theory, it should emit energy continuously. Due to loss of energy, orbits shrink and the electron falls into the nucleus. As a result, the atom should collapse. But in reality atoms are stable.
2. Rutherford model does not explain why electrons stay in fixed orbits and why they do not lose energy continuously.
3. Rutherford model does not describe the arrangement of electrons and number of electrons in each orbit.
4. Rutherford model fails to explain spectral lines.



electron spirals into the nucleus

Stability problem: a revolving electron should lose energy and fall into the nucleus

2.7
Z
Atomic No.

Atomic Number and Mass Number

Atomic Number

The total number of protons present in the nucleus of an atom is called atomic number. It is denoted by **Z**.

Atomic No. = No. of protons (p)

In an atom, the number of protons is same as no. of electrons.

Mass Number

The sum of protons and neutrons present in the nucleus of an atom is called mass number and it is denoted as **A**.

Mass number = No. of protons + No. of neutrons

Q1 **Calculate number of protons, neutrons and electrons in $^{85}_{35}\mathrm{Br}$.**

Solution

Protons = 35, Electrons = 35

Neutrons $= 85 - 35 = 50$

Q2 **The number of electrons, protons and neutrons in a species are equal to 18, 16 and 16 respectively. Assign the proper symbol of the species.**

Solution

$^{32}_{16}\mathrm{S}^{2-}$

Q3 **How many protons and neutrons are there in the following?**

Solution

	Species	Protons	Neutrons
(a)	$^{14}_{6}C$	6	$14 - 6 = 8$
(b)	$^{16}_{8}O$	8	8
(c)	$^{13}_{6}C$	6	$13 - 6 = 7$
(d)	$^{24}_{12}Mg$	12	$24 - 12 = 12$
(e)	$^{56}_{26}Fe$	26	30
(f)	$^{88}_{38}Sr$	38	50

2.8
Is
Isotopes

Isotopes, Isobars and Isotones

Isotopes
Atoms of same element having different mass number are called isotopes.

e.g. $^{1}_{1}H$, $^{2}_{1}H$, $^{3}_{1}H$

Isobars
Atoms of different elements having same mass number are called isobars.

e.g. $^{40}_{18}Ar$, $^{40}_{20}Ca$, $^{40}_{19}K$

Isotones
Atoms of different elements which contain same number of neutrons are called isotones.

e.g. $^{14}_{6}C$ (n = 8), $^{16}_{8}O$ (n = 8), $^{15}_{7}N$ (n = 8)

2.9
Em
EM Waves

Wave Nature of Electromagnetic Radiations

Electromagnetic Wave Theory

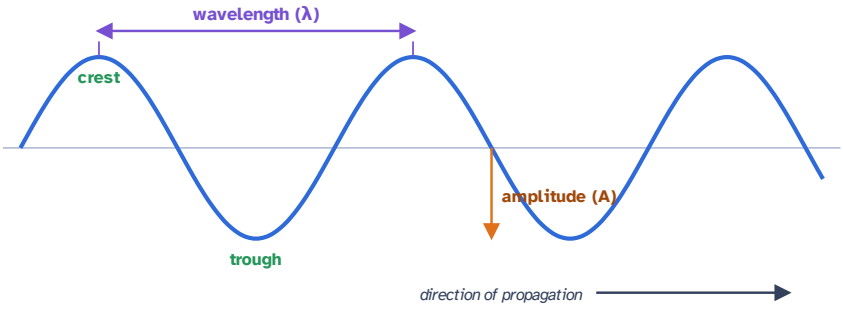
This theory was given by **James Clerk Maxwell in 1870**. According to this theory, light, X-rays, γ-rays etc. travel as electromagnetic waves. These radiations emit energy in the form of radiations called **radiant energy**.

Characteristics of EM waves

1. EM waves emit energy continuously as waves.
2. EM waves are associated with electric and magnetic field.
3. Electric and magnetic fields are perpendicular to each other and also perpendicular to the direction of wave propagation.
4. All electromagnetic waves travel with the velocity of light $(3 \times 10^{8}\ \text{m s}^{-1})$.
5. These rays do not require any medium to propagate.
   e.g. The sun rays reach us after passing through space which has no material medium.

Original notes p. 4

Characteristics of Waves



Crest, trough, wavelength and amplitude of a wave

1 Wavelength (λ)

It is the distance between any two consecutive crests or troughs. It is measured in metre.

2 Frequency (ν)

It is the no. of wavelengths which pass through a point in one second. It is also defined as cycles per second. It is measured in Hertz.

Hz = cycle per second

It is denoted by ν (nu).

3 Amplitude (A)

It is height of crest or depth of trough.

4 Velocity (c)

Velocity of wave is the linear distance travelled by wave in one second. It is measured in $\mathrm{m\ s^{-1}}$ and denoted by **c**.

5 Wave number $(\bar{\nu})$

It is the no. of waves present in one cm length along the direction of propagation. It is denoted by $\bar{\nu}$ and measured in $\mathrm{m^{-1}}$. It is the reciprocal of wavelength.

$$\lambda = \frac{1}{\bar{\nu}} \quad \text{or} \quad \bar{\nu} = \frac{1}{\lambda}$$

Relationship in wavelength, frequency and wave number

$$\nu = \frac{c}{\lambda} \quad \text{or} \quad c = \nu \times \lambda$$

$$\lambda = \frac{1}{\bar{\nu}}, \quad \text{therefore} \quad \nu = c\,\bar{\nu}$$

Electromagnetic Spectrum

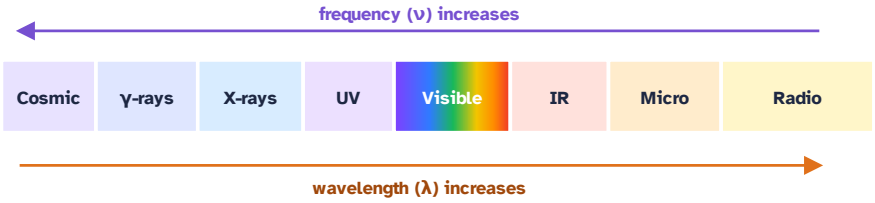
Electromagnetic spectrum

Electromagnetic spectrum is the arrangement of all electromagnetic radiations in order of increasing frequency (ν) or decreasing wavelength (λ).

Order of EM spectrum

Decreasing frequency and increasing wavelength:

Cosmic rays > Gamma rays > X-rays > UV rays > Visible light > IR rays > Micro waves > Radio waves



The electromagnetic spectrum in order of increasing wavelength

Radiations	Wavelength	Frequency	Uses V2
1. Cosmic rays	10^{-16} m	10^{24}	Scientific research
2. Gamma rays	10^{-11} to 10^{-12} m	10^{19} to 10^{20}	Cancer treatment
3. X-rays	10^{-11} to 10^{-8} m	10^{16} to 10^{19}	Medical imaging
4. Ultraviolet ray	10^{-8} to 4×10^{-7} m	10^{14} to 10^{16}	Sterilisation
5. Visible light	4×10^{-7} to 7×10^{-7} m	4×10^{14} to 7.5×10^{14}	Vision
6. Infrared rays	10^{-7} to 10^{-3} m	3×10^{11} to 4.3×10^{14} Hz	Heat, remote ...
7. Micro waves	10^{-3} to 10^{-1} m	10^{9} to 10^{11} Hz	Cooking, radar
8. Radio waves	$> 10^{-1}$ m	10^{9} to 10^{3}	Broadcasting

Visible spectrum

Violet 400 nm	Indigo	Blue	Green	Yellow	Orange	Red 700 nm V3

Human eyes can detect only visible spectrum.

Q1 The Vividh Bharati station of All India Radio broadcasts on a frequency of 1368 kHz. Calculate wavelength of electromagnetic radiations emitted by transmitter. Which part of electromagnetic spectrum does it belong to?

Solution

Frequency, $\nu = 1368 \text{ kHz} = 1368 \times 10^3 \text{ s}^{-1}$ V4

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8 \text{ m s}^{-1}}{1368 \times 10^3 \text{ s}^{-1}}$$

$$\lambda = 0.002193 \times 10^5 \text{ m} = 219.3 \text{ m}$$

$\lambda = 219.3$ m

V5 *The second part of the question is not answered in the original notes.*

Q2 **The wavelength range of the visible spectrum extends from violet (400 nm) to red (750 nm). Express these wavelengths in frequency.**

Solution

Violet: V6

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m s}^{-1}}{4 \times 10^{-7} \text{ m}} = 7.5 \times 10^{14} \text{ s}^{-1}$$

Red:

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m s}^{-1}}{7.5 \times 10^{-7} \text{ m}} = 4 \times 10^{14} \text{ s}^{-1}$$

Violet: 7.5×10^{14} s^{-1} | **Red:** 4×10^{14} s^{-1}

Q3 **Calculate wave number and frequency of yellow radiations having wavelength 5800 Å** $(5800 \times 10^{-10}$ m**).** V7

Solution

$$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{58 \times 10^{-8} \text{ m}} = 0.0172 \times 10^8 \text{ m}^{-1} = 1.72 \times 10^6 \text{ m}^{-1}$$

Frequency:

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m s}^{-1}}{58 \times 10^{-8} \text{ m}} = 0.051 \times 10^{16} \text{ s}^{-1} = 5.1 \times 10^{14} \text{ s}^{-1}$$

$\bar{\nu} = 1.72 \times 10^6$ m^{-1} | $\nu = 5.1 \times 10^{14}$ s^{-1}

Original notes p. 5

2.10 **h** Planck

Planck's Quantum Theory

Particle nature of Electromagnetic Radiations

This theory was proposed by **Max Planck** to explain particle nature of EM radiations. The main features of this theory are:

1. Energy is emitted or absorbed in discrete packets called **quanta** or **photons**.

2. Energy of photon is directly proportional to the frequency.

$$E \propto \nu \qquad E = h\nu$$

where h is Planck's constant $(6.626 \times 10^{-34}$ J s$)$

3 Total amount of energy absorbed or emitted by a body is some whole number quanta.

$$E = nh\nu, \quad n = 1, 2, 3 \ldots$$

4 The energy associated with 1 mole of photons is called **1 Einstein of energy**.

$$E = N_A h\nu = 6.022 \times 10^{23} \frac{hc}{\lambda}$$

Q1 **Calculate the energy of one mole of photons of radiations whose frequency is** 5×10^{14} Hz. $(E = N_A h\nu)$

Solution

$$E = 6.022 \times 10^{23}\ \text{mol}^{-1} \times 5 \times 10^{14}\ \text{s}^{-1} \times 6.626 \times 10^{-34}\ \text{J s}$$

$$E = 199.51 \times 10^{3}\ \text{J mol}^{-1} = 199.51\ \text{kJ mol}^{-1}$$

$E = 199.51\ \text{kJ mol}^{-1}$

Q2 **A 100 watt bulb emits monochromatic light of wavelength 400 nm. Calculate the number of photons emitted per second by the bulb.** $[\lambda = 400 \times 10^{-9}\ \text{m}]$

Solution

Power of bulb = 100 watt $= 100\ \text{J s}^{-1}$

Energy of photon, $E = h\nu = \dfrac{hc}{\lambda}$

$$E = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{4 \times 10^{-7}\ \text{m}} = 4.969 \times 10^{-19}\ \text{J}$$

$$\text{No. of photons} = \frac{100\ \text{J}}{4.969 \times 10^{-19}\ \text{J s}^{-1}} = 20.12 \times 10^{+19}\ \text{s}^{-1} = 2.012 \times 10^{20}\ \text{s}^{-1}$$

$$= 2.012 \times 10^{20}\ \text{Hz}$$

2.012×10^{20} **photons per second**

2.11 Bb Black Body

Black Body Radiations

Black body

A black body is a perfect absorber or emitter of radiation.

Characteristics

1 Absorbs all incident radiations.

2 Emits radiations of all frequencies.

3 Emission depends on temperature only.

e.g. When an iron rod is heated in a furnace it undergoes a series of changes in colour.

Temperature	Colour
Low	Dull Red
Medium	Yellow
High	White
Very High	Blue

As the temperature increases, the intensity of colour increases and frequency also increases.

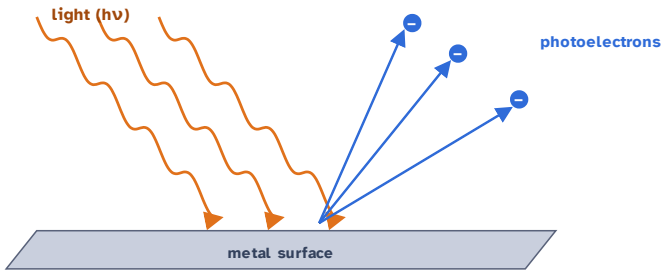
2.12
Pe
Photoelectric

Photoelectric Effect

Photoelectric effect

The emission of electrons from a metal surface when light of suitable frequency falls on it.

Photoelectric effect was discovered by **Heinrich Hertz in 1887** and explained by **Albert Einstein in 1905** by using quantum theory.



Light of suitable frequency ejects electrons from a metal surface

Observations

1. Electrons are emitted immediately when light falls on metal.
2. The no. of electrons ejected is proportional to the intensity/brightness of light.
3. Every metal has a minimum frequency below which photoelectric effect is not observed. This frequency is called **threshold frequency** (ν_0).

Original notes p. 6

Observations (continued)

4. If frequency of incident radiation is greater than threshold frequency i.e. $\nu > \nu_0$, then the electron comes out with certain kinetic energy. Greater the frequency, greater will be the kinetic energy.

Einstein's Explanation

Einstein proposed that light consists of photons, $E = h\nu$. When a photon strikes an electron,

(i) Part of energy of photon used to remove the electron is called **work function** $(W_0 \text{ or } h\nu_0)$.

(ii) Remaining energy becomes kinetic energy.

Einstein equation

$$h\nu = h\nu_0 + \text{K.E.}$$

$$\text{K.E.} = h\nu - h\nu_0 \quad \text{or} \quad \text{K.E.} = h\nu - W_0$$

Work function (W_0) depends upon nature of metal.

Conclusion

(i) If $h\nu < W_0$ → no emission

(ii) If $h\nu = W_0$ → just emission (K.E. = 0) V8

(iii) If $h\nu > W_0$ → electrons emitted with kinetic energy.

Q1 When E.M. radiations of wavelength 300 nm fall on the surface of sodium, the electrons are emitted with a K.E. of 1.68×10^5 J mol^{-1}. What is the minimum energy needed to remove an electron from sodium? What is the maximum wavelength that will cause a photoelectron to be emitted?

Solution

$$\lambda = 300 \times 10^{-9}\ \text{m} = 3 \times 10^{-7}\ \text{m}, \qquad E = \frac{hc}{\lambda}$$

$$E = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{3 \times 10^{-7}\ \text{m}} = 6.626 \times 10^{-19}\ \text{J}$$

$$\text{Energy of one mole of photons} = 6.022 \times 10^{23}\ \text{mol}^{-1} \times 6.626 \times 10^{-19}\ \text{J}$$

$$= 39.9 \times 10^{4}\ \text{J mol}^{-1} = 3.99 \times 10^{5}\ \text{J mol}^{-1}$$

The minimum energy needed to remove one mole of electrons $= h\nu_0 = h\nu - \text{K.E.}$

$$h\nu_0 = 3.99 \times 10^5\ \text{J mol}^{-1} - 1.68 \times 10^5\ \text{J mol}^{-1} = 2.31 \times 10^5\ \text{J mol}^{-1}$$

$$\text{The minimum energy needed to remove one electron} = \frac{2.31 \times 10^5\ \text{J mol}^{-1}}{6.022 \times 10^{23}\ \text{mol}^{-1}}$$

$$= 0.384 \times 10^{-18}\ \text{J} \qquad E = 3.84 \times 10^{-19}\ \text{J}$$

Wavelength: V9

$$\lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{3.84 \times 10^{-19}\ \text{J}} = \frac{19.878}{3.84} \times 10^{-7}\ \text{m}$$

$$= 5.17 \times 10^{-7}\ \text{m} = 517 \times 10^{-9}\ \text{m} = 517\ \text{nm}$$

Minimum energy per electron $= 3.84 \times 10^{-19}$ J | **Maximum wavelength** $= 517$ nm

Q2 The threshold frequency for a metal is 7×10^{14} s^{-1}. Calculate K.E. of an electron emitted when radiation of frequency $\nu = 1 \times 10^{15}$ s^{-1} hits the metal.

Solution

$$\text{K.E.} = \tfrac{1}{2}mv^2 = h(\nu - \nu_0)$$

$$\text{K.E.} = 6.626 \times 10^{-34}\ \text{J s}\,(1 \times 10^{15}\ \text{s}^{-1} - 7 \times 10^{14}\ \text{s}^{-1})$$

$$= 6.626 \times 10^{-34}\ \text{J s}\,(10 \times 10^{14}\ \text{s}^{-1} - 7 \times 10^{14}\ \text{s}^{-1})$$

$$= 6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{14}\ \text{s}^{-1}$$

$$\text{K.E.} = 19.878 \times 10^{-20} = 1.9878 \times 10^{-19}\ \text{J}$$

$\text{K.E.} = 1.9878 \times 10^{-19}\ \text{J}$ V10

2.13
As
Spectra

Atomic Spectra

Atomic spectra

Arrangement of radiations of different wavelengths obtained when light is analysed using a spectroscope is called atomic spectra.

Spectroscope

It is an instrument to separate light into different wavelengths. It contains a prism for dispersion of light and a telescope for observation. If a photographic plate is used instead of telescope, then it is known as a **spectrograph**. V11

Types of Atomic Spectra

① Emission spectra | **② Absorption spectra**

1. Emission spectra

This spectrum is obtained when radiations emitted by a source are passed through a prism and then received on a photographic plate.

Original notes p. 7

Types of Emission spectra

① Continuous spectra

Continuous spectrum has all wavelengths without any gap.

e.g. When sunlight passes through a prism, it splits into seven bands of colours (VIBGYOR). These colours are continuous and are also overlapping each other.

② Line spectrum

Line spectrum consists of sharp, bright lines at specific wavelengths.

e.g. In case of sodium, two yellow lines corresponding to wavelength 589 nm and 589.6 nm are noticed.

Line spectra is also called **atomic spectra** and acts as **fingerprint of elements**.

Continuous, line (emission) and absorption spectra

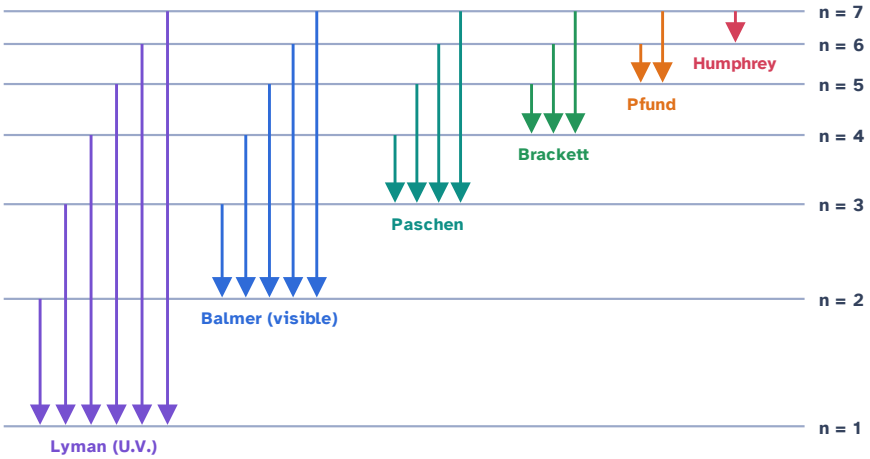
2. Absorption Spectra

When sunlight or white light is passed through the vapours of a substance, and the transmitted light is allowed to strike a prism, dark lines appear in the continuous spectrum. The dark lines represent absorbed wavelengths. This spectrum is called **absorption spectrum**.

2.14
H
H-Spectrum

Line Spectrum of Hydrogen

When an electric discharge is passed through hydrogen gas at low pressure it emits light. When this light is analysed using a spectroscope, it gives discrete bright lines instead of a continuous spectrum. This is called the **line spectrum of hydrogen**. The lines are arranged in different series and each line corresponds to a specific wavelength.



Spectral series of hydrogen: Lyman (U.V.) to n = 1, Balmer (visible) to n = 2, Paschen to n = 3, Brackett to n = 4, Pfund to n = 5, Humphrey to n = 6

Rydberg formula

Rydberg in 1890 gave a simple theoretical equation for the calculation of wavelength of the spectral lines in different series of hydrogen spectrum. This equation is known as Rydberg formula/equation.

$$\frac{1}{\lambda} = \bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

Here, R_H = Rydberg constant $= 109677\ \text{cm}^{-1}$; n_1 = ground state, n_2 = excited state

Q **What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from an energy level n = 4 to an energy level n = 2?**

Solution

Here, $n_1 = 2,\ n_2 = 4$

$$\bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = 109677\ \text{cm}^{-1} \left[\frac{1}{2^2} - \frac{1}{4^2}\right]$$

$$\bar{\nu} = 109677\ \text{cm}^{-1} \left[\frac{1}{4} - \frac{1}{16}\right] = 109677\ \text{cm}^{-1} \left[\frac{4-1}{16}\right]$$

$$\bar{\nu} = 109677\ \text{cm}^{-1} \times \frac{3}{16} = 20564.44\ \text{cm}^{-1}$$

$$\lambda = \frac{1}{\bar{\nu}} = \frac{1}{20564.44\ \text{cm}^{-1}} = 0.0000486\ \text{cm}$$

$$\lambda = 0.0000486 \times 10^{-2}\ \text{m}$$

$$\lambda = 4.86 \times 10^{-7}\ \text{m} = 486 \times 10^{-9}\ \text{m}$$

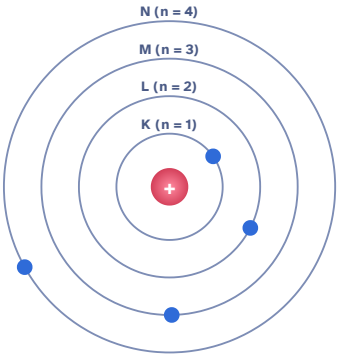
$\lambda = 486\ \text{nm}$

Original notes p. 8

2.15
Bo
Bohr

Bohr's Model of Atom

Bohr's model of atom was proposed by **Niels Bohr in 1913** to overcome the limitations of Rutherford model of atom.



Electrons revolve in fixed circular orbits (energy levels) K, L, M, N

Postulates

1 Quantised Circular Orbits

Electrons revolve around nucleus in fixed circular paths called orbits (shells) which are represented as K, L, M, N or n = 1, 2, 3, 4, ... Each orbit has fixed energy, that's why these are also called **energy levels**.

2 No Energy Loss in Orbit

Electron does not radiate energy while revolving in a fixed orbit. That's why atom is stable.

3 Quantisation of Angular Momentum

Only those orbits are allowed where angular momentum is

$$mvr = \frac{nh}{2\pi}$$

Here, n = orbital no. 1, 2, 3, 4, ...; r = radius, m = mass of electron, v = velocity, h = Planck's constant

4 Energy Emission and Absorption

Electrons jump between orbits. They absorb energy when they move to higher orbits and emit energy when they come to a lower orbit.

$$\text{Energy change, } \Delta E = E_2 - E_1$$

5 Energy of Electron

The important property of an electron is the energy of its stationary state.

$$E_n = -R_H \left[\frac{1}{n^2}\right] \qquad R_H = 2.18 \times 10^{-18}\ \text{J}$$

6 Radius of Orbit (Bohr Radius)

The radius of electron's orbit in H-atom is

$$r_n = \frac{0.529\ \text{Å} \times n^2}{Z}$$

Achievements of Bohr's Model of Atom or Merits

1. Bohr's theory explains the stability of atom.
2. Bohr's theory has been used to derive Balmer's formula. V12

Balmer's formula — derivation

According to Bohr's theory, $E_n = -R_H \left[\frac{1}{n^2}\right]$

1. $$\Delta E = E_2 - E_1, \qquad E_2 = -R_H \left[\frac{1}{n_2^2}\right] \text{ and } E_1 = -R_H \left[\frac{1}{n_1^2}\right]$$
2. $$\therefore\ \Delta E = -R_H \left[\frac{1}{n_2^2}\right] - \left[-R_H \left[\frac{1}{n_1^2}\right]\right] = -R_H \left[\frac{1}{n_2^2}\right] + \left[R_H \left[\frac{1}{n_1^2}\right]\right]$$
3. $$\Delta E = R_H \left[\frac{1}{n_1^2}\right] - R_H \left[\frac{1}{n_2^2}\right] = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$
4. $$\Delta E = 2.18 \times 10^{-18}\ \text{J} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$
5. We know that, $\Delta E = h\nu$
$$\nu = \frac{\Delta E}{h} = \frac{2.18 \times 10^{-18}\ \text{J}}{6.626 \times 10^{-34}\ \text{J s}} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$
6. $$\nu = 0.329 \times 10^{16} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{s}^{-1}$$
$$\nu = 3.29 \times 10^{15} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{s}^{-1}$$
7. $$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{c/\nu} = \frac{\nu}{c}$$

8 $$\bar{\nu} = \frac{\nu}{c} = \frac{3.29 \times 10^{15}\ \text{s}^{-1}}{3 \times 10^{8}\ \text{m s}^{-1}} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

9 $$\bar{\nu} = 1.09677 \times 10^{7}\ \text{m}^{-1} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\bar{\nu} = 1.09677 \times 10^{5}\ \text{cm}^{-1} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

V13 *Line endings cut off in the scan were completed.*

3 Bohr's theory explains atomic spectra of hydrogen.

Limitations

1. Bohr's theory could not explain the spectra of multielectron systems.
2. It fails to explain **Zeeman effect** i.e. the splitting of lines in the magnetic field.
3. It fails to explain **Stark effect** i.e. splitting of lines in the presence of electric field.
4. It could not explain ability of atoms to form molecules.
5. It could not explain dual character of atom.

V14 *Words completed from cut-off line endings are shown with a dotted underline.*

Original notes p. 9

Q What are the frequency and wavelength of a photon emitted during a transition from n = 5 state to n = 2 state in the H-atom?

Solution

$n_1 = 5,\ n_2 = 2$

$$\Delta E = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = -2.18 \times 10^{-18}\ \text{J} \left[\frac{1}{5^2} - \frac{1}{2^2}\right]$$

$$\Delta E = 2.18 \times 10^{-18}\ \text{J} \left[\frac{1}{25} - \frac{1}{4}\right]$$

$$\Delta E = 2.18 \times 10^{-18}\ \text{J} \left[\frac{4 - 25}{100}\right] = 2.18 \times 10^{-18} \times \frac{-21}{100}\ \text{J}$$

$$\Delta E = -4.58 \times 10^{-19}\ \text{J}$$

It is an emission energy. V15

Frequency:

$$\nu = \frac{\Delta E}{h} = \frac{4.58 \times 10^{-19}\ \text{J}}{6.626 \times 10^{-34}\ \text{J s}}$$

$$\nu = 0.691 \times 10^{15}\ \text{s}^{-1} = 6.91 \times 10^{14}\ \text{Hz}$$

Wavelength:

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{6.91 \times 10^{14}\ \text{s}^{-1}}$$

$$\lambda = 0.434 \times 10^{-6}\ \text{m} = 434 \times 10^{-9}\ \text{m} = 434\ \text{nm}$$

$\nu = 6.91 \times 10^{14}\ \text{Hz}$ | $\lambda = 434\ \text{nm}$

Q **Calculate the energy associated with the first orbit of He^+. What is the radius of this orbit?**

Solution

$$E_n = -\frac{2.18 \times 10^{-18}\ \text{J} \times Z^2}{n^2}$$

For He^+-ion, $n = 1,\ Z = 2$

$$E_n = \frac{-2.18 \times 10^{-18}\ \text{J} \times (2)^2}{(1)^2} = -2.18 \times 10^{-18} \times 4\ \text{J}$$

$$E_n = -8.72 \times 10^{-18}\ \text{J}$$

Radius:

$$r_n = \frac{0.529\ \text{Å} \times n^2}{Z} = \frac{0.529 \times 10^{-10}\ \text{m} \times (1)^2}{2}$$

$$r_n = 0.2645 \times 10^{-10}\ \text{m} = 0.2645\ \text{Å}$$

$E_1 = -8.72 \times 10^{-18}\ \text{J}$ | $r_1 = 0.2645\ \text{Å}$

Distinction between Particle and Wave

Particle	Wave
① Localised position in space	① Delocalised in space
② Particles do not interfere with each other	② They can interfere with each other
③ Particle can be counted	③ Waves cannot be counted

2.16 λ de Broglie

Dual Behaviour of Matter (De-Broglie's Equation)

In **1924, Louis de-Broglie**, a French physicist, advanced a new idea by stating that all moving particles have dual nature i.e. wave and particle nature. He stated that a wave is associated with every moving particle and is known as **matter wave**.

De-Broglie gave a relationship between wavelength and momentum of microscopic particles. This relationship is known as De-Broglie's wave relation. According to De-Broglie wave relation, the momentum of any microscopic particle is inversely proportional to the wavelength of the wave associated with it.

De-Broglie's wave relation

$$\lambda \propto \frac{1}{p} \Rightarrow \lambda = \frac{h}{p} \Rightarrow \lambda = \frac{h}{mv} \qquad [p = mv]$$

Here, λ = wavelength, p = momentum, m = mass, v = velocity

Derivation

1. According to Planck's Quantum theory, the energy of photon is

$$E = h\nu \qquad \text{— (i)}$$

2. According to Einstein equation, the energy of photon is

$$E = mc^2 \quad \text{— (ii)}$$

3. Comparing equations (i) and (ii):

$$h\nu = mc^2 \Rightarrow \frac{hc}{\lambda} = mc^2$$

4. $$\frac{h}{\lambda} = mc \Rightarrow \frac{h}{mc} = \lambda \Rightarrow \lambda = \frac{h}{mc}$$

5. $$\lambda = \frac{h}{mv} \Rightarrow \lambda = \frac{h}{p} \qquad [c = v]$$

$$\lambda \propto \frac{1}{p}$$

This equation is known as **De-Broglie's wave equation** and λ is known as **De-Broglie's wavelength**.

Significance of De-Broglie's wave equation

1. It explains wave-particle duality.
2. It is helpful in calculating wavelength.
3. It helps in the calculation of velocity of particles.

Original notes p. 10

Q **What will be the wavelength of a ball of mass 0.1 kg moving with a velocity of 10 m s^{-1}?**

Solution

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J s}}{0.1\ \text{kg} \times 10\ \text{m s}^{-1}} = 6.626 \times 10^{-34}\ \text{J m}^{-1}\text{s}^2\text{kg}^{-1}$$

$$\lambda = 6.626 \times 10^{-34}\ \text{kg m}^2\text{s}^{-2}\ \text{m}^{-1}\text{s}^2\ \text{kg}^{-1} \qquad [\text{J} = \text{kg m}^2\text{s}^{-2}]$$

$$\lambda = 6.626 \times 10^{-34}\ \text{m}$$

Q **A golf ball has a mass of 40 kg and a speed of 45 m s^{-1}. If speed can be measured with accuracy of 2%, calculate the uncertainty in position.**

Solution

Uncertainty in speed is 2%, i.e.

$$45 \times \frac{2}{100} = 0.9\ \text{m s}^{-1}$$

Apply,

$$\Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.626 \times 10^{-34}\ \text{J s}}{4 \times 3.14 \times 40\ \text{kg} \times 0.9\ \text{m s}^{-1}}$$

$$\Delta x = \frac{6.626 \times 10^{-34}}{452.16}\ \text{J s kg}^{-1}\text{m}^{-1}\text{s}$$

$$\Delta x = 0.0146 \times 10^{-34}\ \text{kg m}^2\text{s}^{-2}\ \text{kg}^{-1}\text{m}^{-1}\text{s}^2$$

$$\Delta x = 1.46 \times 10^{-36}\ \text{m}$$

2.17 Δ Heisenberg

Heisenberg Uncertainty Principle

Heisenberg Uncertainty Principle was proposed by **Werner Heisenberg in 1927**.

Statement

This principle states that it is impossible to determine simultaneously the exact position and momentum of a microscopic particle.

Alternatively, the principle may also be stated as: the product of uncertainty in position and uncertainty in momentum of a microscopic particle is always constant and is equal to or greater than $h/4\pi$.

Mathematically

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

Here, Δx is uncertainty in position and Δp is the uncertainty in momentum.

$$\Delta p = m \times \Delta v$$

Conditions

(i) If Δx is very small, then Δp is large, i.e. if uncertainty in position is small (certainty in position is large) then uncertainty in momentum is large. In this case, the position of a microscopic particle can be measured accurately and momentum cannot be measured accurately.

(ii) If Δp is very small, then momentum can be measured accurately.

Explanation

Let us consider an electron. In order to measure its position, the radiations of shorter wavelength and higher energy are used. In order to measure the momentum of electron, the radiations of longer wavelength and smaller energy are used. V16

Significance

1. Exact position and momentum cannot be known simultaneously.
2. It forms basis of quantum mechanics.
3. Effect negligible for macroscopic objects.
4. Electron does not move in fixed orbits (no definite path). V17

Q A microscope using suitable photons is employed to locate an electron in an atom within a distance of 0.1 Å. What is the uncertainty involved in the measurement of its velocity? V18

Solution

$\Delta x = 0.1 \times 10^{-10}$ m, $\quad m = 9.11 \times 10^{-31}$ kg

$$\Delta v = \frac{h}{4\pi \times \Delta x \times m} = \frac{6.626 \times 10^{-34}\ \text{J s}}{4 \times 3.14 \times 0.1 \times 10^{-10}\ \text{m} \times 9.11 \times 10^{-31}\ \text{kg}}$$

$$\Delta v = \frac{6.626 \times 10^{-34}\ \text{J s}}{11.44 \times 10^{-41}\ \text{m kg}} = 0.579 \times 10^{7}\ \text{J s m}^{-1}\text{kg}^{-1}$$

$$\Delta v = 5.79 \times 10^{6}\ \text{kg m}^{2}\text{s}^{-2}\ \text{s m}^{-1}\text{kg}^{-1} = 5.79 \times 10^{6}\ \text{m s}^{-1}$$

$\Delta v = 5.79 \times 10^{6}\ \text{m s}^{-1}$

2.18 Ψ Q.M. Model

Quantum Mechanical or Wave Mechanical Model of Atom

Quantum Mechanical Model of atom was developed by **Erwin Schrödinger in 1926** and it is based on ideas of Werner Heisenberg.

This model explains the behaviour of electrons using wave mechanics. Electron is not a simple particle only, it also behaves like a wave.

Original notes p. 11

Main Postulates

1. **Wave Nature of Electron:** Electrons show dual nature (particle + wave).
2. Electrons do not move in fixed orbits. Instead, they exist in orbitals.
3. **Orbitals:** Orbitals are the regions where probability of finding an electron is maximum.
4. **Schrödinger wave equation:** It is a mathematical equation that describes the wave nature and energy of an electron in an atom and gives the probability of finding the electron at a particular position. It is a differential equation.

Schrödinger wave equation

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}[E - V]\psi = 0$$

Here x, y, z are space co-ordinates

m = mass of electron, E = total energy of e^-

ψ (Psi) → wave function of electron. It is also called **Eigen-function**.

$\frac{\partial^2\psi}{\partial x^2}$ = second differential of ψ with respect to x axis and so on.

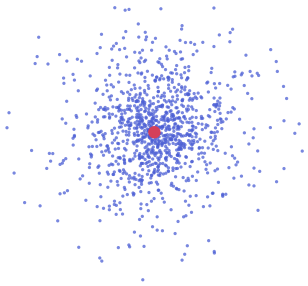
The solution of Schrödinger wave equation gives the value of E and ψ.

5. **Wave function (ψ):** ψ has no direct physical meaning but its square i.e. ψ^2 gives probable electron density i.e. atomic orbital. This leads to **electron cloud concept**.
6. Exact position and momentum of electron cannot be known together.

Atomic Orbital

Atomic orbital

Atomic orbital is the space around the +vely charged nucleus where the probability of finding an electron is maximum. The probable electron density is not 100% and it decreases as the distance from the nucleus increases and never becomes zero even at large distance from the nucleus. The probability of finding an electron around nucleus is 90–95%.



Cloud representation of electron

Difference between Orbit and Orbital

Orbit	Orbital
① An orbit is a well defined circular path around nucleus in which electron revolves.	① It is the space around nucleus where probability of finding an electron is maximum.
② It represents the movement of an electron around the nucleus in one plane.	② It represents the movement of electron in three dimensions.
③ In an orbit, the position as well as momentum of the electron can be known with certainty.	③ In an orbital, the position as well as momentum of electron cannot be known with certainty.
④ All orbits are circular.	④ Orbitals have different shapes.
⑤ The maximum number of electrons in an orbit is given by formula $2n^2$.	⑤ An orbital cannot accommodate more than two electrons.

2.19
Qn
Q. Numbers

Quantum Numbers

Quantum numbers

Quantum numbers are those numbers which provide complete information about electrons. These are:

- ① **Principal Quantum number**
- ② **Azimuthal Quantum number**
- ③ **Magnetic Quantum number**
- ④ **Spin Quantum number**

1 Principal Quantum number

It is represented by letter **n**. It provides the following information:

(i) It tells us about main energy shell. For K-shell, n = 1; for L-shell, n = 2; for M-shell, n = 3; for N-shell, n = 4 and so on.

(ii) It gives the value of average distance of an electron from nucleus. $r_n = \frac{0.529\ \text{Å} \times n^2}{Z}$

(iii) It gives the value of average energy of an electron in a shell. $E_n = -R_H \frac{Z^2}{n^2}$

(iv) The total no. of electrons in a shell is given by formula $2n^2$.

Original notes p. 12

2 Azimuthal Quantum number or Angular momentum Quantum number

It was given by **Sommerfeld** and denoted by l. It provides the following information:

(i) Number of subshells present within any main energy level

(ii) Relative energy of the subshells belonging to the same shell

(iii) Shape of subshells

(iv) Orbital angular momentum (mvr) of electron

$$mvr = \frac{h}{2\pi}\sqrt{l(l+1)}$$

The value of l can be derived by using formulae $l = (n-1),\ (n-2),\ (n-3)\ldots$

For K-shell, $n = 1$:
$l = n - 1 = 1 - 1 = 0$

For L-shell, $n = 2$:
$l = n - 1 = 2 - 1 = 1$
$l = n - 2 = 2 - 2 = 0$

For M-shell, $n = 3$:
$l = n - 1 = 3 - 1 = 2$
$l = n - 2 = 3 - 2 = 1$
$l = n - 3 = 3 - 3 = 0$

For N-shell, $n = 4$:
$l = n - 1 = 4 - 1 = 3$
$l = n - 2 = 4 - 2 = 2$
$l = n - 3 = 4 - 3 = 1$
$l = n - 4 = 4 - 4 = 0$

Value of n	Value of l	Name of subshell	No. of subshells
1	0	1s	one
2	0, 1	2s, 2p	two
3	0, 1, 2	3s, 3p, 3d	three
4	0, 1, 2, 3	4s, 4p, 4d, 4f	four

3 Magnetic Quantum number

Magnetic Q. No. was proposed by **Linde**. Magnetic Q. No. is denoted by m_l. It specifies the orientations (directions) of an orbital in space. It tells us how the orbital is aligned in the presence of magnetic field. The value of m_l depends on the azimuthal Q. No. and is calculated by using formula,

$$m_l = -l \ \ldots \ 0 \ \ldots \ +l$$

For s-subshell, $l = 0$	$m_l = 0$	i.e. s-subshell contains one orbital
For p-subshell, $l = 1$	$m = -1, 0, +1$	three orbitals
For d-subshell, $l = 2$	$m = -2, -1, 0, +1, +2$	⑤ orbitals
For f-subshell, $l = 3$	$m = -3, -2, -1, 0, +1, +2, +3$	⑦ orbitals

4 Spin Quantum number

It is denoted by m_s. This number gives the information about direction of spinning of electron present in any orbital. Spin Q. No. has two possible values: V19

$m_s = +\frac{1}{2}$ **for clockwise spin** | $m_s = -\frac{1}{2}$ **for anticlockwise spin**

Radial Wave Function (R)

Radial wave function describes how the probability of finding an electron varies with distance from the nucleus (r).

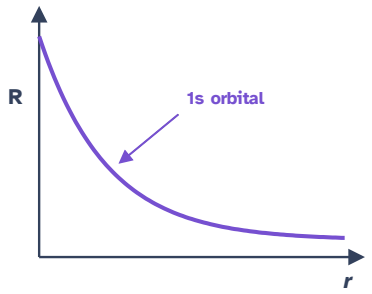
Radial wave function depends only on distance of the electron from the nucleus. It depends upon the quantum numbers n and l.

Radial wave function determines the **size of orbital**.

Radial wave function gives the radial probability density.

Formula

$$P(r) = 4\pi r^2 [R(r)]^2$$



R versus r for 1s orbital

Radial nodes

Radial nodes are points (or spherical regions) where the probability of finding an electron is zero at a certain distance from nucleus.

Formula: No. of radial nodes $= n - l - 1$ V20

Importance of Radial wave function

(i) It helps in determining orbital size. V21

(ii) Explains distribution of electrons in shells.

(iii) *[left blank in the original notes]*

Original notes p. 13

Angular Wave Function

Angular wave function describes the angular position (direction) of the electron in an atom. It depends upon

θ (theta) → polar angle	ϕ (phi) → azimuthal angle

Angular wave function is often denoted as $Y(\theta, \phi)$. V22

It is the part of the electronic wave function that describes the angular shape and spatial orientation of an atomic orbital. It depends upon quantum numbers l and m and is quite independent of n.

Angular node $= l$

2.20
spd
Shapes

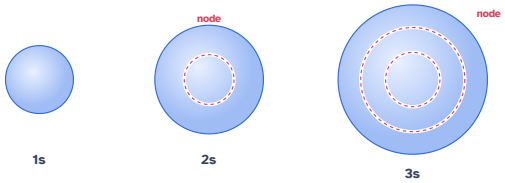
Shapes of Atomic Orbitals

Shape of s-orbital

The s-orbital is present in s-subshell for which $l = 0$ and $m = 0$. s-orbital with only one orientation has a **spherical shape** with uniform electron density along all three axes. 1s orbital is smaller in size than 2s-orbital which in turn is smaller than 3s-orbital and so on.

The region where probability of finding an electron is almost zero is called **nodal surface or node**. The no. of nodes is calculated by formula $(n - 1)$.

1s orbital has no node, 2s orbital has one node and 3s-orbital has two nodes.

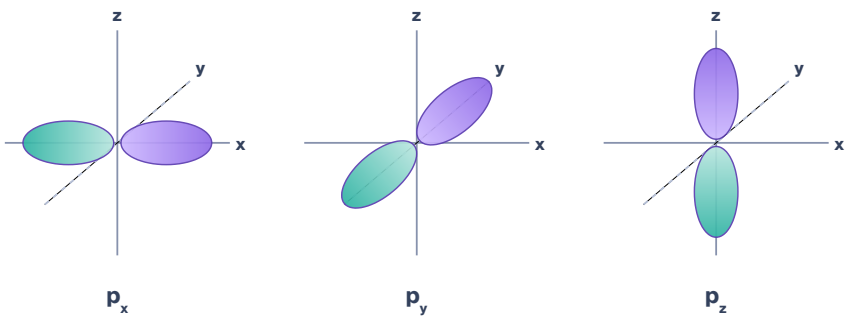


1s, 2s and 3s orbitals with nodes

Shape of p-orbitals

p-orbitals are present in p-subshells for which $l = 1$ and m can have three values i.e. three possible orientations –1, 0, +1. Thus, there are three orbitals in the p-subshell which are designated as p_x, p_y, p_z orbitals depending on the axis along which they are directed. Each orbital consists of two lobes. The probability of finding an electron in both the lobes is the same. The p-orbitals have **dumb-bell shape** because the lobes are similar to a pair of dumb-bells.

p-orbitals are directional in nature and hence influence the shape of molecule. Orbitals belonging to the same energy shell have same energy. These are **degenerate orbitals**.

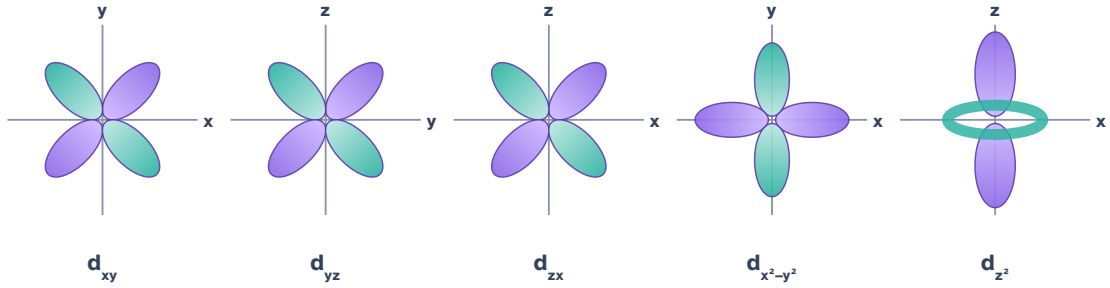


p_x, p_y and p_z orbitals

Shape of d-orbitals

d-orbitals are present in d-subshell for which $l = 2$ and m can have five values i.e. five possible orientations –2, –1, 0, +1, +2. Thus d-subshell has five orbitals designated as $d_{xy}, d_{yz}, d_{zx}, d_{x^2-y^2}$ and d_{z^2}. These are degenerate orbitals.

The first four orbitals have similar shape and the fifth d-orbital i.e. d_{z^2} is different in shape. The first four orbitals have **clover leaf shape**. In the fourth orbital, the lobes are directed along x and y-axis. d_{z^2} has **doughnut shape**.



The five d-orbitals

Original notes p. 14

2.21 Af Filling

Filling of Orbitals in an Atom — Electronic Configuration

1. Aufbau Principle

Aufbau is a German word which means *'building up'*.

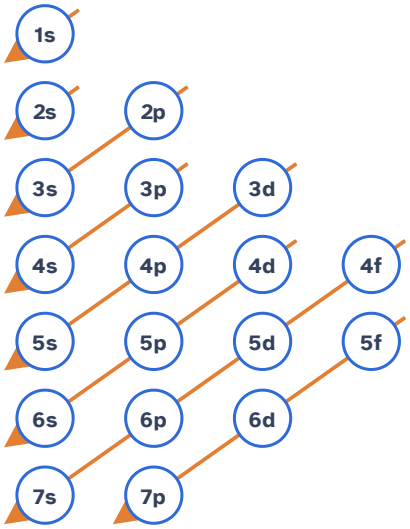
Aufbau Principle

According to Aufbau Principle, the electrons fill orbitals in order of increasing energies of orbitals, i.e. lowest energy orbitals are filled first. The order of energy is decided by the **(n + l) rule**.

(n + l) rule

The orbital with lower (n + l) value is filled first. If two orbitals have the same (n + l) value, then the orbital with lower n value fills first.

The sequence for filling orbitals is:



Sequence of filling of orbitals

4s versus 3d

4s-orbital: $n + l = 4 + 0 = 4$

3d-orbital: $n + l = 4 + 2 = 5$ V23

That's why 4s is filled first and 3d is filled later.

2. Pauli's Exclusion Principle

Pauli's Exclusion Principle

According to Pauli's Exclusion Principle, no two electrons in an atom can have the same set of four quantum numbers. Two electrons present in an atom can have a maximum of three quantum numbers the same.

Shell	Subshell	Orbital	Electrons	n	l	m	s
K	s	s	↑	1	0	0	$+\frac{1}{2}$
			↓	1	0	0	$-\frac{1}{2}$

3. Hund's Multiplicity Rule

Hund's rule

According to this rule, when more than one orbital of equal energy is available, the electrons will first occupy these orbitals singly with parallel spins. The pairing of electrons will then take place.

This is due to the reason that electrons carry same charge and when two electrons with same spin are filled in one orbital, they repel each other. In order to minimize the repulsion, the electrons move as far apart as possible. V24

d^5 | ↑ | ↑ | ↑ | ↑ | ↑ | Correct

d^5 | ↑↓ | ↑↓ | ↑ | | | Wrong V25

Stability of Completely Filled and Half-Filled Orbitals

① Completely filled orbital

When an orbital is completely filled, it becomes extra stable.

Example: s-subshell is stable when it has 2 electrons (s^2); p-subshell is stable when it has 6 electrons (p^6); d-subshell is stable when it has 10 electrons (d^{10}).

Reason:

1. Symmetrical distribution of electrons
2. Minimum repulsion
3. Large exchange of energies

② Half-filled orbitals

When a subshell is exactly half filled, it is also more stable. *e.g.* d-subshell → 5 electrons

Reason:

1. Maximum number of unpaired electrons
2. Large exchange of energies
3. Symmetrical distribution of electrons

Examples

Chromium: 24 → $[\mathrm{Ar}]\, 4s^1\, 3d^5$

Copper: 29 → $[\mathrm{Ar}]\, 4s^1\, 3d^{10}$

Original notes p. 15

2.22 Ex Exercise I

Solved Exercise Questions

Q1 **(i) Calculate the number of electrons which will together weigh 1 gram. (ii) Calculate the mass and charge of one mole of electrons.** V26

Solution

(i) Mass of an e^- $= 9.11 \times 10^{-28}$ g

9.11×10^{-28} g is the mass of 1 electron

$$1 \text{ gram is the mass of } \frac{1 \text{ g}}{9.11 \times 10^{-28} \text{ g}} e^- \text{s}$$

$$= 0.1098 \times 10^{28} e^- \text{s} = 1.098 \times 10^{27} e^- \text{s}$$

(ii) Mass of 1 e^- $= 9.11 \times 10^{-31}$ kg

$$\text{Mass of 1 mole of electrons} = 6.022 \times 10^{23} \times 9.11 \times 10^{-31} \text{ kg}$$

$$= 54.8 \times 10^{-8} \text{ kg} = 5.48 \times 10^{-7} \text{ kg}$$

Charge on 1 e^- $= 1.602 \times 10^{-19}$ C

$$\text{Charge on 1 mole of } e^-\text{s} = 1.602 \times 10^{-19} \times 6.022 \times 10^{23} = 9.65 \times 10^4 \text{ C}$$

(i) 1.098×10^{27} **electrons | (ii)** 5.48×10^{-7} kg, 9.65×10^4 C

Q2 **(i) Calculate the total number of electrons present in one mole of** CH_4**.**
(ii) Find (a) total number and (b) the total mass of neutrons in 7 mg of ^{14}C**. (Assume mass of neutron** $= 1.675 \times 10^{-27}$ kg**)**
(iii) Find (a) total no. and (b) the total mass of protons in 34 mg of NH_3 **at STP. Will the answer change if the temperature and pressure are changed?**

Solution

(i) 1 mole of $CH_4 = 6.022 \times 10^{23}$ CH_4 molecules

No. of e^-s in $CH_4 = 6 + 4 = 10$

$$\text{No. of } e^-\text{s in 1 mole of } CH_4 = 6.022 \times 10^{23} \times 10 = 6.022 \times 10^{24} e^-\text{s}$$

(ii) (a) Calculation of total no. of carbon atoms: Mass of carbon $= 7$ mg $= 7 \times 10^{-3}$ g

Atomic mass of $^{14}C = 14$ g mol^{-1}

14 g of carbon contain 6.022×10^{23} C-atoms

1 g of carbon contain $\frac{6.022 \times 10^{23}}{14}$ C-atoms

$$7 \text{ mg of carbon atoms contain } \frac{6.022 \times 10^{23}}{14} \times 7 \times 10^{-3} \text{ g carbon atoms}$$

$$= 0.5 \times 6.022 \times 10^{20} = 3.011 \times 10^{20} \text{ C atoms}$$

(b) Total number of neutrons and mass of neutrons:

Total neutrons in $^{14}C = 14 - 6 = 8n$

$$\text{Total neutrons in 7 mg of C-atoms} = 8 \times 3.011 \times 10^{20} n = 24.088 \times 10^{20} = 2.4088$$

V27 *The power of ten after 2.4088 is not written in the original.*

$$\text{Mass of } 24.088 \times 10^{20}\ n = 24.088 \times 10^{20} \times 1.675 \times 10^{-27}\ \text{kg}$$
$$= 40.34 \times 10^{-7}\ \text{kg} = 4.034 \times 10^{-6}\ \text{kg}$$

(iii) (a) Calculation of total no. of NH_3 molecules

Gram molecular mass of $NH_3 = 17\ \text{g mol}^{-1}$

17 g of $NH_3 = 6.022 \times 10^{23}$ NH_3 molecules

1 g of $NH_3 = \dfrac{6.022 \times 10^{23}}{17}$ NH_3 molecules

$$34 \times 10^{-3}\ \text{g of } NH_3 = \frac{6.022 \times 10^{23}}{17} \times 34 \times 10^{-3}\ \text{g } NH_3$$
$$= 12.044 \times 10^{20}\ NH_3 \text{ molecules} = 1.2044 \times 10^{21}$$

(b) Total p^+ and mass of total p^+

No. of protons in $NH_3 = 10$

$$\text{No. of protons in 34 g of } NH_3 = 10 \times 1.2044 \times 10^{21} = 1.2044 \times 10^{22}\ p$$

Mass of 1 proton $= 1.67 \times 10^{-27}$ kg

$$\text{Mass of } 1.2044 \times 10^{22} \text{ protons} = 1.67 \times 10^{-27} \times 1.2044 \times 10^{22} = 2.01 \times 10^{-5}\ \text{kg}$$

(i) $6.022 \times 10^{24}\ e^-$ **| (ii)** 3.011×10^{20} **C atoms;** 4.034×10^{-6} kg **| (iii)** 1.2044×10^{22} **protons;** 2.01×10^{-5} kg

V28 *The last part of (iii) is not answered in the original notes.*

Original notes p. 16

Q3 How many protons and neutrons are present in the following nuclei? (i) $^{13}_{6}C$ (ii) $^{16}_{8}O$ (iii) $^{24}_{12}Mg$ (iv) $^{56}_{26}Fe$ (v) $^{88}_{38}Sr$

Solution

	Nucleus	Protons	Neutrons
(i)	$^{13}_{6}C$ (Atomic No. = 6)	6	7
(ii)	$^{16}_{8}O$	8	8
(iii)	$^{24}_{12}Mg$	12	12
(iv)	$^{56}_{26}Fe$	26	30
(v)	$^{88}_{38}Sr$	38	50

Q4 Write the complete symbol for the atom (X) with the given atomic no. (Z) and atomic mass (A).

Solution

(i) Z = 17, A = 35 — $^{35}_{17}Cl$

(ii) Z = 92, A = 233 — $^{233}_{92}U$

(iii) Z = 4, A = 9 — $^{9}_{4}Be$

Q5 **Yellow light emitted from a sodium lamp has a wavelength (λ) of 580 nm. Calculate the frequency (ν) and wave number $(\bar{\nu})$ of yellow light.**

Solution

$$\text{Frequency, } \nu = \frac{c}{\lambda} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{580 \times 10^{-9}\ \text{m}} = 5.17 \times 10^{-14}\ \text{s}^{-1}$$

V29 *Exponent reproduced as it appears in the notes — please verify.*

$$\text{Wave No. } (\bar{\nu}) = \frac{1}{\lambda} = \frac{1}{580 \times 10^{-9}\ \text{m}} = 1.724 \times 10^{6}\ \text{m}^{-1}$$

$\bar{\nu} = 1.724 \times 10^{6}\ \text{m}^{-1}$

Q6 **Calculate the energy of each photon which (i) corresponds to light of frequency 3×10^{15} Hz; (ii) has wavelength of 0.50 Å.** V30

Solution

$$\text{(i)}\ E = h\nu = 6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{15}\ \text{s}^{-1}$$

$$= 19.878 \times 10^{-19}\ \text{J} = 1.9878 \times 10^{-18}\ \text{J}$$

$$\text{(ii)}\ E = h\nu = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{0.50 \times 10^{-10}\ \text{m}}$$

$$= \frac{19.878 \times 10^{-26}\ \text{J}}{0.50 \times 10^{-10}} = 39.756 \times 10^{-16}\ \text{J} = 3.9756 \times 10^{-15}\ \text{J}$$

(i) 1.9878×10^{-18} J | **(ii)** 3.9756×10^{-15} J

Q7 **Calculate the wavelength, frequency and wave number of a light wave whose period is 2.0×10^{-10} s .** V31

Solution

$$\nu = \frac{1}{T} = \frac{1}{2.0 \times 10^{-10}\ \text{s}} = 0.5 \times 10^{10}\ \text{s}^{-1} = 5 \times 10^{9}\ \text{s}^{-1}$$

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{5 \times 10^{9}\ \text{s}^{-1}} = 0.6 \times 10^{-1} = 6 \times 10^{-2}\ \text{m}$$

$$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{6.0 \times 10^{-2}\ \text{m}} = 0.166 \times 10^{2}\ \text{m}^{-1} = 1.66 \times 10^{1}\ \text{m}^{-1}$$

$\nu = 5 \times 10^{9}\ \text{s}^{-1}$ | $\lambda = 6 \times 10^{-2}\ \text{m}$ | $\bar{\nu} = 1.66 \times 10^{1}\ \text{m}^{-1}$

Q8 **What is the no. of photons of light with wavelength 4000 pm which provide 1 joule of energy?**

Solution

Energy of photon $(E) = \frac{hc}{\lambda}$

$$E = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3.0 \times 10^{8}\ \text{m s}^{-1}}{4000 \times 10^{-12}\ \text{m}}$$

$$E = \frac{19.878 \times 10^{-14}}{4000} = \frac{19.878 \times 10^{-17}}{4} = 4.969 \times 10^{-17}\ \text{J}$$

4.969 is the energy of 1 photon

$$\therefore \text{ 1 joule is the energy of photons} = \frac{1}{4.969 \times 10^{-17}\ \text{J}} = 0.2012 \times 10^{17} = 2.012 \times 10^{16}\ \text{photons}$$

2.012×10^{16} photons V32

Original notes p. 17

Q9 **A photon of wavelength 4×10^{-7} m strikes on metal surface; the work function of the metal being 2.13 eV. Calculate (i) the energy of photon, (ii) the kinetic energy of the emission, (iii) the velocity of photoelectron. (Given 1 eV $= 1.6020 \times 10^{-19}$ J)** V33

Solution

(i) The energy of photon

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{4 \times 10^{-7}\ \text{m}}$$

$$E = 4.969 \times 10^{-19}\ \text{J} = \frac{4.969 \times 10^{-19}\ \text{J} \times 1\ \text{eV}}{1.602 \times 10^{-19}\ \text{J}}$$

$$E = 3.1\ \text{eV}$$

(ii) Kinetic energy of emission

Apply, $h\nu = h\nu_0 + \text{K.E.}$

$$\text{K.E.} = h\nu - h\nu_0 \qquad [h\nu_0 = W_0 \text{ work function}]$$

$$\text{K.E.} = E - W_0 = 3.1\ \text{eV} - 2.13\ \text{eV} = 0.97\ \text{eV}$$

(iii) Velocity of photoelectron

$$\text{K.E.} = \tfrac{1}{2}mv^2 = 0.97\ \text{eV} \qquad v^2 = \frac{0.97\ \text{eV} \times 2}{m}$$

$$v^2 = \frac{2 \times 0.97 \times 1.602 \times 10^{-19}\ \text{J}}{9.11 \times 10^{-31}\ \text{kg}} = 0.34 \times 10^{12}\ \text{J/kg}$$

$$v^2 = 0.34 \times 10^{12}\ \text{kg m}^2\text{s}^{-2}\,\text{kg}^{-1} \qquad v^2 = 0.34 \times 10^{12}\ \text{m}^2\text{s}^{-2}$$

$$v = \sqrt{0.34 \times 10^{12}\ \text{m}^2\text{s}^{-2}} = 0.583 \times 10^{6}\ \text{m s}^{-1}$$

$$v = 0.583 \times 10^{6}\ \text{m s}^{-1} = 5.83 \times 10^{5}\ \text{m s}^{-1}$$

(i) 3.1 eV | (ii) 0.97 eV | (iii) $5.83 \times 10^{5}\ \text{m s}^{-1}$

Q10 **Electromagnetic radiation of wavelength 242 nm is just sufficient to ionise the sodium atom. Calculate the ionisation energy of sodium in kJ mol^{-1}.**

Solution

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{242 \times 10^{-9}\ \text{m}}$$

$$E = \frac{19.878}{242} \times 10^{-17}\ \text{J} = 0.0821 \times 10^{-17}\ \text{J}$$

$$\text{Ionisation energy per mole} = 0.0821 \times 10^{-17}\ \text{J} \times 6.022 \times 10^{23}\ \text{mol}^{-1}$$

$$= 0.494 \times 10^{6} = 494 \times 10^{3}\ \text{J mol}^{-1} = 494\ \text{kJ mol}^{-1}$$

Ionisation energy $= 494\ \text{kJ mol}^{-1}$

Q11 **A 25 watt bulb emits monochromatic yellow light of wavelength 0.57 µm. Calculate the rate of emission of quanta per second.**

Solution

Energy of 1 photon $(E) = \dfrac{hc}{\lambda}$

$$E = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{0.57 \times 10^{-6}\ \text{m}}$$

$$E = \frac{19.878}{0.57} \times 10^{-20} = 34.87 \times 10^{-20} = 3.487 \times 10^{-19}\ \text{J}$$

$$\text{Rate of emission of quanta per second} = \frac{25\ \text{Watt}}{\text{Energy}} = \frac{25\ \text{J s}^{-1}}{3.487 \times 10^{-19}\ \text{J}} = 7.17 \times 10^{19}\ \text{s}^{-1}$$

7.17×10^{19} **quanta per second**

Q12 **Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength 6800 Å. Calculate threshold energy (ν_0) and work function (W_0) of the metal.**

Solution

$$\text{Threshold frequency } (\nu_0) = \frac{c}{\lambda} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{6800 \times 10^{-10}\ \text{m}}$$

$$\nu_0 = \frac{3 \times 10^{8}\ \text{s}^{-1}}{68 \times 10^{-8}} = 0.0441 \times 10^{16}\ \text{s}^{-1} = 4.41 \times 10^{14}\ \text{s}^{-1}$$

$$\text{Work function, } W_0 = h\nu_0 = 6.626 \times 10^{-34}\ \text{J s} \times 4.41 \times 10^{14}\ \text{s}^{-1}$$

$$= 29.2 \times 10^{-20}\ \text{J} = 2.92 \times 10^{-19}\ \text{J}$$

$\nu_0 = 4.41 \times 10^{14}\ \text{s}^{-1}$ | $W_0 = 2.92 \times 10^{-19}\ \text{J}$

Q13 **What is the wavelength of the light emitted when the electron in a hydrogen atom undergoes transition from the energy level with n = 4 to energy level n = 2? What is the colour corresponding to this wavelength? (Given $R_H = 109678\ \text{cm}^{-1}$)**

Solution

$$\bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{cm}^{-1}; \qquad n_1 = 2,\ n_2 = 4$$

$$\bar{\nu} = 109678\ \text{cm}^{-1} \left[\frac{1}{2^2} - \frac{1}{4^2}\right] = 109678\ \text{cm}^{-1} \left[\frac{1}{4} - \frac{1}{16}\right]$$

$$\bar{\nu} = 109678\ \text{cm}^{-1} \times \frac{16 - 4}{64} = 109678\ \text{cm}^{-1} \times \frac{12}{64}$$

$$\bar{\nu} = 20564.28\ \text{cm}^{-1}$$

$$\lambda = \frac{1}{\bar{\nu}} = \frac{1}{20564.28\ \text{cm}^{-1}} = 0.0000486 = 486 \times 10^{-7}\ \text{cm}^{-1}$$

$$= 486 \times 10^{-9}\ \text{m} = 486\ \text{nm}$$

V34 *Values reproduced exactly as written.*

$\lambda = 486\ \text{nm}$ | **Colour: Blue green**

Original notes p. 18

Q14 **How much energy is required to ionise a hydrogen atom if an electron occupies n = 5 orbit? Compare your answer with the ionisation energy of H-atom (energy required to remove the electron from n = 1 orbit).**

Solution

Energy for a hydrogen electron present in a particular energy shell,

$$E_n = -\frac{13.12 \times 10^5}{n^2}\ \text{J mol}^{-1}$$

$$E_n = -\frac{1312}{n^2}\ \text{kJ} = -\frac{1312 \times 10^3\ \text{J}}{n^2}\ \text{per mol}$$

$$E_n = \frac{-1312 \times 10^3}{n^2 \times 6.022 \times 10^{23}}\ \text{J atom}^{-1}$$

$$E_n = \frac{-218.6 \times 10^3\ \text{J atom}^{-1}}{n^2 \times 10^{23}} = -218.6 \times 10^{-20} = -2.186 \times 10^{-18}\ \text{J atom}^{-1}$$

Step I: Ionisation energy for hydrogen electron present in orbit n = 5

$$IE_5 = E_\infty - E_5 = 0 - \left[-\frac{2.18 \times 10^{-18}}{n^2}\right]$$

$$IE_5 = \frac{2.18 \times 10^{-18}}{25}\ \text{J atom}^{-1} = 0.0872 \times 10^{-18}\ \text{J atom}^{-1} = 8.72 \times 10^{-20}\ \text{J atom}^{-1}$$

Step II:

$$IE_1 = \frac{2.18 \times 10^{-18}}{1}\ \text{J atom}^{-1} = 2.18 \times 10^{-18}\ \text{J atom}^{-1}$$

On comparing:

$$\frac{IE_1}{IE_2} = \frac{2.18 \times 10^{-18}}{8.72 \times 10^{-20}} = \frac{0.25}{10^{-2}} = 25$$

V35 *Ratio label reproduced as written.*

The energy required to remove an electron from the first orbit in a hydrogen atom is 25 times the energy needed to remove an electron from the 5th orbit.

Q15 **What is the maximum number of emission lines when the excited electron of a hydrogen atom in n = 6 drops to the ground state?** V36

Solution

$$\text{The maximum no. of emission lines} = \frac{n(n-1)}{2} = \frac{6(6-1)}{2} = \frac{6 \times 5}{2} = 15$$

15 emission lines

Q16 **(i) The energy associated with the first orbit in hydrogen atom is $-2.17 \times 10^{-18}\ \text{J atom}^{-1}$. What is the energy associated with the fifth orbit? (ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.** V37

Solution

(i) Energy of first orbit $(E_1 = -2.17 \times 10^{-18}\ \text{J atom}^{-1})$

For 5th orbit:

$$E_5 = \frac{-2.17 \times 10^{-18}\ \text{J atom}^{-1}}{(5)^2} = \frac{-2.17 \times 10^{-18}}{25} = -0.0868 \times 10^{-18}$$

$$E_5 = -8.68 \times 10^{-20}\ \text{J atom}^{-1}$$

(ii) Radius of 5th orbit:

$$r_5 = \frac{0.529\ \text{Å} \times n^2}{Z} = \frac{0.529 \times 10^{-10}\ \text{m} \times (5)^2}{1}$$

$$r_5 = 0.529 \times 10^{-10}\ \text{m} \times 25$$

$$r_5 = 13.225 \times 10^{-10}\ \text{m} = 13.225\ \text{Å}$$

(i) $E_5 = -8.68 \times 10^{-20}\ \text{J atom}^{-1}$ **| (ii)** $r_5 = 13.225 \times 10^{-10}\ \text{m}$

Q17 Calculate the wave number for the longest wavelength transition in the Balmer series of atomic hydrogen.

Solution

For Balmer series, $n_1 = 2$. Longest wavelength means lowest energy. Therefore this transition occurs between $n_1 = 2$ to $n_2 = 3$.

$$\text{Apply, } \bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = 109677\ \text{cm}^{-1} \left[\frac{1}{2^2} - \frac{1}{3^2}\right]$$

$$\bar{\nu} = 109677\ \text{cm}^{-1} \left[\frac{1}{4} - \frac{1}{9}\right] = 109677\ \text{cm}^{-1} \left[\frac{9-4}{36}\right]$$

$$\bar{\nu} = 109677\ \text{cm}^{-1} \times \frac{5}{36} = 15232.9\ \text{cm}^{-1}$$

$$= 1.52 \times 10^{-4}\ \text{cm}^{-1} = 1.52 \times 10^{6}\ \text{m}^{-1}$$

V38 *Exponent reproduced as written — please verify.*

$\bar{\nu} = 15232.9\ \text{cm}^{-1} = 1.52 \times 10^{6}\ \text{m}^{-1}$

Original notes p. 19

Q18 What is the energy in joules required to shift the electron of the hydrogen atom from first Bohr orbit to the fifth orbit and what is the wavelength of light emitted when the electron returns to the ground state? The ground state electronic energy is 2.18×10^{-11} **erg.** V39

Solution

Energy required to shift electron from n = 1 to n = 5

$$E_1 = -2.18 \times 10^{-18}\ \text{J}$$

$$E_5 = \frac{-2.18 \times 10^{-18}\ \text{J}}{5^2} = \frac{-2.18 \times 10^{-18}\ \text{J}}{25} = -0.0872 \times 10^{-18} = 8.72 \times 10^{-20}\ \text{J}$$

$$\Delta E = E_5 - E_1 = -8.72 \times 10^{-20} - [-2.18 \times 10^{-18}]$$

$$= -0.0872 \times 10^{-18} + 2.18 \times 10^{-18} = (2.18 - 0.0872) \times 10^{-18}\ \text{J}$$

$$\Delta E = 2.09 \times 10^{-18}\ \text{J}$$

$$\Delta E = h\nu \Rightarrow \Delta E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{\Delta E}$$

$$\lambda = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{2.09 \times 10^{-18}\ \text{J}} = \frac{19.878}{2.09} \times 10^{-8}\ \text{m}$$

$$\lambda = 9.50 \times 10^{-8}\ \text{m} = 95.0 \times 10^{-9}\ \text{m} = 950 \times 10^{-10} = 950\ \text{Å}$$

$\Delta E = 2.09 \times 10^{-18}\ \text{J}$ | $\lambda = 950\ \text{Å}$

Q19 **The electronic energy in hydrogen atom is given by $E_n = (-2.18 \times 10^{-18})/n^2$ J. Calculate the energy required to remove an electron completely from n = 2 orbit. What is the longest wavelength of light in cm that can be used to cause this transition?**

Solution

$$\Delta E = E_\infty - E_2 = 0 - \left[\frac{-2.18 \times 10^{-18}\ \text{J}}{2^2}\right]$$

$$\Delta E = \frac{2.18 \times 10^{-18}}{4}\ \text{J} = 0.545 \times 10^{-18}\ \text{J} = 5.45 \times 10^{-19}\ \text{J}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{5.45 \times 10^{-19}\ \text{J}} = 3.644 \times 10^{-7}\ \text{m}$$

$\lambda = 3.644 \times 10^{-5}\ \text{cm}$

Q20 **Calculate the wavelength of an electron moving with a velocity of $2.05 \times 10^7\ \text{m s}^{-1}$.**

Solution

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J s}}{9.11 \times 10^{-31}\ \text{kg} \times 2.05 \times 10^{7}\ \text{m s}^{-1}}$$

$$\lambda = 0.354 \times 10^{-10}\ \text{J s}^2\,\text{m}^{-1}\,\text{kg}^{-1}$$

$$\lambda = 3.54 \times 10^{-11}\ \text{kg m}^2\text{s}^{-2} \times \text{s}^2\,\text{m}^{-1}\,\text{kg}^{-1}$$

$\lambda = 3.54 \times 10^{-11}\ \text{m}$

Q21 **The mass of an electron is 9.1×10^{-31} kg. If its kinetic energy is 3.0×10^{-25} J, calculate its wavelength.**

Solution

$$\text{K.E.} = \tfrac{1}{2}mv^2 = \tfrac{1}{2} \times 9.1 \times 10^{-31}\ \text{kg} \times v^2$$

$$3 \times 10^{-25}\ \text{J} = \tfrac{1}{2} \times 9.1 \times 10^{-31}\ \text{kg}\ v^2$$

$$\text{or, } v^2 = \frac{3 \times 10^{-25} \times 2\ \text{J}}{9.1 \times 10^{-31}\ \text{kg}}$$

$$v^2 = 0.659 \times 10^{6}\ \text{kg}^{-1}\text{m}^2\text{s}^{-2}\,\text{kg}$$

$$v = \sqrt{65.9 \times 10^4\ \text{m}^2\text{s}^{-2}} = \sqrt{(8.12 \times 10^2\ \text{m s}^{-1})^2}$$

$$v = 8.12 \times 10^2\ \text{m s}^{-1}$$

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J s}}{9.1 \times 10^{-31}\ \text{kg} \times 8.12 \times 10^2\ \text{m s}^{-1}}$$

$$\lambda = 0.08967 \times 10^{-5}\ \text{kg m}^2\text{s}^{-2}\,\text{m}^{-1}\,\text{s}^2$$

$\lambda = 8967\ \text{Å}$

Q22 **Which of the following are isoelectronic species?** Na^+, K^+, Mg^{2+}, Ca^{2+}, S^{2-}, Ar

Solution

Na^+ and Mg^{2+} are isoelectronic (10 e^-s)

K^+, Ca^{2+}, S^{2-}, Ar are isoelectronic (18 e^-s)

Q23 **(i) Write the electronic configuration of the following ions: (a) H^- (b) Na^+ (c) O^{2-} (d) F^-**

Solution

(a) H^-	$1s^2$
(b) Na^+	$1s^2\,2s^2\,2p^6$
(c) O^{2-}	$1s^2\,2s^2\,2p^6$
(d) F^-	$1s^2\,2s^2\,2p^6$

Original notes p. 20

Q23 **(continued) (ii) What are the atomic numbers of elements whose outermost electronic configurations are represented by: (a) $3s^1$ (b) $2p^3$ (c) $3d^6$?**

Solution

(a) $3s^1$	Sodium (Z = 11)
(b) $2p^3$	Nitrogen (Z = 7)
(c) $3d^6$	Iron (Z = 26)

(iii) Which atoms are indicated by the following configurations?

(a) $[He]\,2s^1$	Li (Z = 3)
(b) $[Ne]\,3s^2\,3p^3$	P (Z = 15)
(c) $[Ar]\,4s^2\,3d^1$	Sc (Z = 21)

Q24 **What is the lowest value of n which allows g orbitals to exist?**

Solution

n = 5

Q25 **An electron is in one of the 3d-orbitals. Give the possible values of n, l and m_l for the electron.**

Solution

$n = 3,\ l = 2,\ m_l = -2, -1, 0, +1, +2$

Q26 **An atom of an element contains 29 electrons and 35 neutrons. Deduce (i) the number of protons and (ii) the electronic configuration of the element.**

Solution

No. of protons = No. of electrons = 29

Electronic configuration $= [\mathrm{Ar}]\, 4s^1\, 3d^{10}$

Q27 **Give the number of electrons in the species:** H_2^+, H_2 **and** O_2^+

Solution

$H_2^+ = 1e^-$ | $H_2 = 2e^-$ | $O_2^+ = 15e^-$

Q28 **(i) An atomic orbital has n = 3. What are the possible values of** l **and** m_l**?** V40

Solution

$l = 0, 1, 2$

m_l: for $l = 0,\ m_l = 0$; for $l = 1,\ m_l = -1, 0, +1$; for $l = 2,\ m_l = -2, -1, 0, +1, +2$

(ii) List the quantum numbers (m_l **and** l**) of electron in 3d orbital.**

n = 3; 3d orbital: $n = 3,\ l = 2,\ m_l$ can have any of the values $-2, -1, 0, +1, +2$

(iii) Which of the following orbitals are possible? 1p, 2s, 2p and 3f

1p and 3f are not possible.

Q29 **Using s, p and d notations, describe the orbitals with the following quantum numbers: (a) n = 1,** l **= 0 (b) n = 4,** l **= 3 (c) n = 3,** l **= 0 (d) n = 4,** l **= 2 (e) n = 3,** l **= 1**

Solution

(a)	1s-orbital
(b)	4f-orbital
(c)	3s-orbital
(d)	4d-orbital
(e)	3p-orbital

Q30 **From the following sets of quantum numbers, state which are possible. Explain why the others are not possible.**

Solution

	Set	Answer
(i)	$n = 0,\ l = 0,\ m_l = 0,\ m_s = +\frac{1}{2}$	Not possible, ∵ n cannot be zero
(ii)	$n = 1,\ l = 0,\ m_l = 0,\ m_s = -\frac{1}{2}$	Possible
(iii)	$n = 1,\ l = 1,\ m_l = 0,\ m_s = +\frac{1}{2}$	Not possible, because for n = 1, l cannot be equal to 1
		Original notes p. 21
(iv)	$n = \ldots,\ l = 0,\ m_l = +1,\ m_s = +\frac{1}{2}$ V41	Not possible because for $l = 0$, m_l cannot be +1. It must be zero.
(v)	$n = 3,\ l = 3,\ m_l = -3,\ m_s = +\frac{1}{2}$	Not possible because for n = 3, $l \neq 3$
(vi)	$n = 3,\ l = 1,\ m_l = 0,\ m_s = +\frac{1}{2}$	Possible

V42 *Line endings cut off in the scan were completed.*

Q31 **How many electrons in an atom may have the following quantum numbers? (a) n = 4, $m_s = -\frac{1}{2}$ (b) n = 3, $l = 0$**

Solution

(a) For n = 4, total e^-s $= 2n^2 = 2(4)^2 = 32\, e^-$s

∴ Total electrons with $m_s = -\frac{1}{2}$ is 16 e^-s.

(b) n = 3, l = 0: Two electrons (3s)

(a) 16 electrons | (b) 2 electrons (3s)

Q32 **Show that the circumference of the Bohr orbit for the hydrogen atom is an integral multiple of the de Broglie wavelength associated with the electron revolving around the orbit.**

Solution

According to Bohr's theory

$$mvr = \frac{nh}{2\pi} \quad \text{or} \quad 2\pi r = \frac{nh}{mv} \quad \text{or,}\ mv = \frac{nh}{2\pi r} \quad -(\text{i})$$

According to de Broglie's equation

$$\lambda = \frac{h}{mv} \quad \text{or} \quad mv = \frac{h}{\lambda} \quad -(\text{ii})$$

From (i) and (ii):

$$\frac{nh}{2\pi r} = \frac{h}{\lambda} \quad \text{or} \quad \frac{n}{2\pi r} = \frac{1}{\lambda}$$

$$n\lambda = 2\pi r$$

Thus, circumference of Bohr's orbit is an integral multiple of de Broglie's wavelength.

Q33 **What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n = 4 to n = 2 of He^+ spectrum?**

Solution

$$\bar{\nu} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

For He^+-ion, Z = 1, $n_1 = 2,\ n_2 = 4$ V43

$$\bar{\nu} = R_H \times 2^2 \left[\frac{1}{2^2} - \frac{1}{4^2}\right] = R_H \times 4 \left[\frac{16-4}{64}\right]$$

$$\bar{\nu} = R_H \times \frac{4 \times 12}{64} = R_H \times \frac{3}{4} \quad \text{— (i)}$$

For hydrogen spectrum:

$$\bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \quad \text{— (ii)}$$

From (i) and (ii):

$$R_H \frac{3}{4} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \Rightarrow \frac{3}{4} = \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

This corresponds to $n_1 = 1$ and $n_2 = 2$: the transition has taken place in the Lyman series from n = 2 to n = 1.

Q34 **Calculate the energy required for the process $He^+(g) \longrightarrow He^{2+}(g) + e^-$. The ionisation energy for the H atom in the ground state is 2.18×10^{-18} J atom^{-1}.**

Solution

$$E_n = \frac{2.18 \times 10^{-18} \times Z^2}{n^2} \text{ J atom}^{-1}$$

For H atom $= 2.18 \times 10^{-18}$ J atom^{-1}

$$\text{For } He^+ \text{ ion } (Z = 2),\ E_n = \frac{2.18 \times 10^{-18} \times (2)^2}{(1)^2}$$

$E_n = 8.72 \times 10^{-18}$ J atom^{-1}

Q35 **2×10^8 atoms of carbon are arranged side by side. Calculate the radius of carbon atom if the length of this arrangement is 2.4 cm.**

Solution

Length of arrangement = 2.4 cm; Total no. of C-atoms present $= 2 \times 10^8$

$$\text{Diameter of each C-atom} = \frac{2.4}{2 \times 10^8} = 1.2 \times 10^{-8} \text{ cm}$$

$$\text{Radius of each carbon atom} = \frac{1.2 \times 10^{-8}}{2} \text{ cm} = 0.6 \times 10^{-8} = 6 \times 10^{-9} \text{ cm}$$

Radius $= 6 \times 10^{-9}$ cm

Original notes p. 22

Q36 **The diameter of a carbon atom is 0.15 nm. Calculate the number of carbon atoms which can be placed side by side in a straight line across a length of scale of length 20 cm.**

Solution

$$\text{Length of scale} = 20 \text{ cm} = 20 \times 10^{-2} \text{ m} = 20 \times 10^{-2} \times 10^9 \text{ nm} = 20 \times 10^7 \text{ nm} = 2 \times 10^8 \text{ nm}$$

Diameter of carbon atom = 0.15 nm

$$\therefore \text{ No. of carbon atoms placed side by side} = \frac{2 \times 10^8 \text{ nm}}{0.15 \text{ nm}} = 1.33 \times 10^9$$

1.33×10^9 carbon atoms

Q37 The diameter of zinc atom is 2.6 Å. Calculate (i) radius of Zn atom in pm, (ii) no. of atoms present in a length of 1.6 cm if the zinc atoms are arranged side by side lengthwise.

Solution

$$\text{Radius of zinc atom} = \frac{2.6 \times 10^{-10} \text{ m}}{2} = 1.3 \times 10^{-10} \text{ m} = 130 \times 10^{-12} \text{ m}$$

Length of scale $= 1.6 \text{ cm} = 1.6 \times 10^{10}$ pm; Diameter of zinc atom = 260 pm

$$\therefore \text{ No. of zinc atoms placed side by side} = \frac{1.6 \times 10^{10} \text{ pm}}{260 \text{ pm}} = 6.154 \times 10^7$$

(i) 130 pm | (ii) 6.154×10^7 atoms

Q38 A certain particle carries 2.5×10^{-16} C of static electric charge. Calculate the number of electrons present in it.

Solution

Magnitude of charge $(q) = 2.5 \times 10^{-16}$ C; Charge on 1 electron $(e) = 1.602 \times 10^{-19}$ C

$$\text{No. of electrons present} = \frac{2.5 \times 10^{-16}}{1.602 \times 10^{-19}} = 1.56 \times 10^3 = 1560 \text{ electrons}$$

1560 electrons

Q39 In Millikan's experiment, the static electric charge on the oil droplet was obtained to be -1.282×10^{-18} C. Calculate the number of electrons present in it. V44

Solution

$$\text{Number of } e^-\text{s} = \frac{-1.282 \times 10^{-18}}{-1.602 \times 10^{-19}} = 0.8 \times 10 = 8 \text{ electrons}$$

8 electrons

Q40 In Rutherford's experiment, generally the thin foil of heavy atoms like gold, platinum etc. have been used to be bombarded by the α-particles. If a thin foil of light atoms like aluminium etc. is used, what difference would be observed from the above results?

Solution

Using a light metal like aluminium will show much smaller deflection effects compared to heavy metals like gold, because the nucleus is less massive and has lower positive charge.

V45 *Line endings cut off in the scan were completed.*

Q41 **Symbols $^{79}_{35}\mathrm{Br}$ and $^{79}\mathrm{Br}$ can be written, whereas symbols $^{35}_{79}\mathrm{Br}$ and $^{35}\mathrm{Br}$ are not accepted. Answer in brief.**

Solution

Correct: $^{79}_{35}\mathrm{Br}$ and $_{35}\mathrm{Br}$ V46

Because, 35 = Atomic No. (Protons) and 79 = Mass number (Protons + Neutrons)

Incorrect: $^{35}_{79}\mathrm{Br}$ and $^{35}\mathrm{Br}$ are not accepted because the atomic no. of the element cannot be 79.

Atomic number is fixed for an element but mass can vary.

Q42 **An element with mass number 81 contains 31.7% more neutrons compared to protons. Assign the symbol to the element.**

Solution

Mass No. A = 81, Neutrons are 31.7% more than protons.

Original notes p. 23

Let no. of protons = x

$$\text{No. of neutrons} = x + x \times \frac{31.7}{100} = x + 0.317x$$

$$\text{Mass No.} = \text{protons} + \text{neutrons}: \quad 81 = x + x + 0.317x \Rightarrow 81 = 2.317x \Rightarrow x = \frac{81}{2.317} = 35$$

No. of protons = 35, No. of neutrons = 81 – 35 = 46; Atomic No. = No. of protons = 35

Symbol $= ^{81}_{35}\mathrm{Br}$

Q43 **An ion with mass number 37 possesses one unit of negative charge. If the ion contains 11.1% more neutrons than electrons, find the symbol of the ion.**

Solution

Let no. of electrons of the ion = x; No. of protons = x – 1

$$\text{No. of neutrons} = x + x \times \frac{11.1}{100} = x + 0.111x = 1.111x$$

$$\text{Mass No.} = x - 1 + 1.111x: \quad 37 = 2.111x - 1 \Rightarrow 38 = 2.111x \Rightarrow x = \frac{38}{2.111} = 18$$

No. of e^-s = 18, No. of protons = 18 – 1 = 17; Atomic No. = 17

Symbol $= ^{37}_{17}\mathrm{Cl}^-$

Q44 **An ion with mass number 56 contains 3 units of positive charge and 30.4% more neutrons than electrons. Assign symbol of the ion.**

Solution

Let no. of electrons in the ion = x; No. of protons = x + 3

$$\text{No. of neutrons} = x + \frac{30.4}{100}x = 1.304x$$

$$\text{Mass No.}, \; 56 = x + 3 + 1.304x = 2.304x + 3 \Rightarrow 56 - 3 = 2.304x \Rightarrow x = \frac{53}{2.304} = 23$$

Atomic No. = 23 + 3 = 26

Symbol $= {}^{56}_{26}\mathrm{Fe}^{3+}$

Q45 **Arrange the following types of radiations in increasing order of wavelength: (a) radiation from microwave oven (b) amber light from traffic signal (c) radiation from FM radio (d) cosmic rays from outer space (e) X-rays**

Solution

Cosmic rays < X-rays < Amber colour < Micro waves < FM

Q46 **Nitrogen laser produces a radiation of wavelength 337.1 nm. If the no. of photons emitted is** 5.6×10^{24}**, calculate the power of this laser.**

Solution

$$\text{Power of laser} = E = Nh\nu = \frac{Nhc}{\lambda}$$

$$E = \frac{5.6 \times 10^{24} \times 6.626 \times 10^{-34}\ \text{J s} \times 3.0 \times 10^{8}\ \text{m s}^{-1}}{337.1 \times 10^{-9}\ \text{m}}$$

$E = 0.33 \times 10^{7}\ \text{J} = 3.3 \times 10^{6}\ \text{J}$

Q47 **Neon gas is generally used in sign boards. If it emits strongly at 616 nm, calculate (a) frequency of emission, (b) distance travelled by this radiation in 30 s, (c) energy of quantum, (d) no. of quanta present if it produces 2 J of energy.**

Solution

$$\text{(i) Frequency, } \nu = \frac{c}{\lambda} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{616 \times 10^{-9}\ \text{m}} = 4.87 \times 10^{14}\ \text{s}^{-1}$$

$$\text{(ii) Distance travelled in 30 s} = \text{Velocity} \times \text{Time} = 3 \times 10^{8}\ \text{m s}^{-1} \times 30\ \text{s} = 9 \times 10^{9}\ \text{m}$$

$$\text{(iii) Energy of quanta} = h\nu = 6.626 \times 10^{-34}\ \text{J s} \times 4.87 \times 10^{14}\ \text{s}^{-1} = 32.27 \times 10^{-20}\ \text{J}$$

$$\text{(iv) No. of quanta in 2 J} = \frac{2\ \text{J}}{32.27 \times 10^{-20}\ \text{J}} = 0.062 \times 10^{20} = 6.2 \times 10^{18}\ \text{photons}$$

$4.87 \times 10^{14}\ \text{s}^{-1}$ | $9 \times 10^{9}\ \text{m}$ | $32.27 \times 10^{-20}\ \text{J}$ | 6.2×10^{18} **photons**

Original notes p. 24

Q48 **In astronomical observations, signals observed from the distant stars are generally weak. If the photon detector receives a total of** 3.15×10^{-18} J **from the radiations of 600 nm, calculate the no. of photons received by the detector.**

Solution

Total energy received $= 3.15 \times 10^{-18}$ J; $\lambda = 600\ \text{nm} = 6 \times 10^{-7}\ \text{m}$

$$\text{Energy of 1 photon} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{6 \times 10^{-7}\ \text{m}} = 3.313 \times 10^{-19}\ \text{J}$$

$$\text{No. of photons} = \frac{\text{Total } E}{E \text{ of one photon}} = \frac{3.15 \times 10^{-18}\ \text{J}}{3.313 \times 10^{-19}\ \text{J}} = 10\ \text{photons}$$

10 photons

Q49 **Lifetimes of the molecules in the excited states are often measured by using pulsed radiation source of duration nearly in the nanosecond range. If the radiation source has the duration of 2 ns and the number of photons emitted during the pulse source is 2.5×10^{15}, calculate the energy of the source.**

Solution

Time duration $= 2 \times 10^{-9}$ s

$$\text{Frequency } (\nu) = \frac{1}{t} = \frac{1}{2 \times 10^{-9}\ \text{s}} = 5 \times 10^{8}\ \text{s}^{-1}$$

$$\text{Energy of 1 photon, } E = h\nu = 6.626 \times 10^{-34}\ \text{J s} \times 5 \times 10^{8}\ \text{s}^{-1} = 33.13 \times 10^{-26}\ \text{J} = 3.313 \times 10^{-25}\ \text{J}$$

No. of photons $= 2.5 \times 10^{15}$

$$\therefore\ \text{Energy of source} = 3.313 \times 10^{-25}\ \text{J} \times 2.5 \times 10^{15} = 8.28 \times 10^{-10}\ \text{J}$$

8.28×10^{-10} J

Q50 **The longest wavelength doublet absorption transition is observed at 589 nm and 589.6 nm. Calculate frequency of each transition and energy difference between two excited states.** V47

Solution

$\lambda_1 = 589\ \text{nm} = 589 \times 10^{-9}\ \text{m}$

$$\nu_1 = \frac{c}{\lambda_1} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{589 \times 10^{-9}\ \text{m}} = 5.0934 \times 10^{14}\ \text{s}^{-1}$$

$\lambda_2 = 589.6\ \text{nm} = 589.6 \times 10^{-9}\ \text{m}$

$$\nu_2 = \frac{c}{\lambda_2} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{589.6 \times 10^{-9}\ \text{m}} = 5.088 \times 10^{14}\ \text{s}^{-1}$$

$$\Delta E = E_1 - E_2 = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2} = hc\left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right]$$

$$\Delta E = 6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}\left[\frac{1}{589} - \frac{1}{589.6}\right]\frac{1}{10^{-9}\ \text{m}}$$

$$\Delta E = \frac{19.878 \times 10^{-26}\ \text{J m}}{10^{-9}\ \text{m}}\left[\frac{589.6 - 589}{589 \times 589.6}\right]$$

$$\Delta E = \frac{19.878 \times 10^{-17} \times 0.6}{347274.4} = 3.43 \times 10^{-22}\ \text{J}$$

$\nu_1 = 5.0934 \times 10^{14}\ \text{s}^{-1}$ | $\nu_2 = 5.088 \times 10^{14}\ \text{s}^{-1}$ | $\Delta E = 3.43 \times 10^{-22}$ J

Q51 **The work function for caesium atom is 1.9 eV. Calculate (a) the threshold wavelength and (b) the threshold frequency of the radiation. If the caesium element is irradiated with a wavelength 500 nm, calculate the kinetic energy and the velocity of the ejected photoelectron.**

Solution

$$E_0 = h\nu_0 = 1.9\ \text{eV} = 1.9 \times 1.602 \times 10^{-19}\ \text{J}$$

$$\text{Threshold frequency} = \nu_0 = \frac{E_0}{h} = \frac{1.9 \times 1.602 \times 10^{-19}\ \text{J}}{6.626 \times 10^{-34}\ \text{J s}}$$

$$\nu_0 = 0.459 \times 10^{15}\ \text{s}^{-1} = 4.59 \times 10^{14}\ \text{s}^{-1}$$

$$\text{Threshold wavelength, } \lambda_0 = \frac{c}{\nu_0} = \frac{3 \times 10^8}{4.59 \times 10^{14}}$$

$$\lambda_0 = 0.6536 \times 10^{-6}\ \text{m} = 653.6 \times 10^{-9}\ \text{m}$$

$$h\nu = h\nu_0 + \text{K.E.} \quad \text{or} \quad E = E_0 + \tfrac{1}{2}mv^2$$

$$\tfrac{1}{2}mv^2 = E - E_0 = hc\left[\frac{1}{\lambda} - \frac{1}{\lambda_0}\right] = 6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^8\ \text{m s}^{-1}\,[\ldots]$$

$$= \frac{19.878 \times 10^{-26}\ \text{J m}}{10^{-9}\ \text{m}}\left[\frac{654 - 500}{654 \times 500}\right] = 19.878 \times 10^{-17}\ \text{J} \times \frac{154}{327000}$$

Original notes p. 25

$$\tfrac{1}{2}mv^2 = 0.00936 \times 10^{-17}\ \text{J}$$

$$v^2 = \frac{9.36 \times 10^{-20}\ \text{J} \times 2}{m} = \frac{18.72 \times 10^{-20}\ \text{J}}{9.1 \times 10^{-31}\ \text{kg}} = 2.057 \times 10^{11}\ \text{kg m}^2\text{s}^{-2}/\text{kg}$$

$$v = \sqrt{20.57 \times 10^{10}\ \text{m}^2\text{s}^{-2}} = \sqrt{(4.5356 \times 10^5\ \text{m s}^{-1})^2}$$

$\lambda_0 = 653.6$ nm | $\nu_0 = 4.59 \times 10^{14}\ \text{s}^{-1}$ | **K.E.** $= 9.36 \times 10^{-20}$ J | $v = 4.5356 \times 10^5\ \text{m s}^{-1}$

V48 *Partly cut-off lines are marked.*

Q52 Following results are observed when sodium metal is irradiated with different wavelengths. Calculate threshold wavelength.

Solution

λ (nm)	500	450	400
$v \times 10^5\ \text{m s}^{-1}$	2.55	4.35	5.35

Let threshold wavelength $= \lambda_0\ \text{nm} = \lambda_0 \times 10^{-9}\ \text{m}$

According to photoelectric effect: $h(\nu - \nu_0) = \tfrac{1}{2}mv^2$, $\quad hc\left[\frac{1}{\lambda} - \frac{1}{\lambda_0}\right] = \tfrac{1}{2}mv^2$

$$\text{(a)}\ \frac{hc}{10^{-9}\ \text{m}}\left[\frac{1}{500} - \frac{1}{\lambda_0}\right] = \tfrac{1}{2}m(2.55 \times 10^5)^2 \quad \text{— (i)}$$

$$\text{(b)}\ \frac{hc}{10^{-9}\ \text{m}}\left[\frac{1}{450} - \frac{1}{\lambda_0}\right] = \tfrac{1}{2}m(4.35 \times 10^5)^2 \quad \text{— (ii)}$$

$$\text{(c)}\ \frac{hc}{10^{-9}\ \text{m}}\left[\frac{1}{400} - \frac{1}{\lambda_0}\right] = \tfrac{1}{2}m(5.35 \times 10^5)^2 \quad \text{— (iii)}$$

Divide eq. (ii) by (i):

$$\frac{\lambda_0 - 450}{450\,\lambda_0} \times \frac{500\,\lambda_0}{\lambda_0 - 500} = \frac{(4.35 \times 10^5)^2}{(2.55 \times 10^5)^2}$$

$$\frac{\lambda_0 - 450}{\lambda_0 - 500} = \frac{(4.35)^2}{(2.55)^2} \times \frac{450}{500} = \frac{18.9}{6.5} \times \frac{45}{50} = \frac{850.5}{325} = 2.62$$

$$\lambda_0 - 450 = 2.62(\lambda_0 - 500) = 2.62\lambda_0 - 1310$$

$$-450 + 1310 = 2.62\lambda_0 - \lambda_0 \Rightarrow 860 = 1.62\lambda_0 \Rightarrow \lambda_0 = \frac{860}{1.62} = 530.8$$

$\lambda_0 = 530.8$ nm

Q53 **The ejection of the photoelectrons from the silver metal in the photoelectric effect experiment can be stopped by applying the voltage of 0.35 V when the radiation 256.7 nm is used. Calculate work function for silver metal.**

Solution

$\lambda = 256.7 \times 10^{-9}$ m, K.E. = 0.35 eV

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{256.7 \times 10^{-9}\ \text{m}}$$

$$E = 0.0774 \times 10^{-17}\ \text{J} = \frac{0.0774 \times 10^{-17}}{1.602 \times 10^{-19}}\ \text{eV} = 0.0483 \times 10^{2} = 4.83\ \text{eV}$$

According to photoelectric effect: $E = E_0 + \text{K.E.}$

$$4.83\ \text{eV} = E_0 + 0.35\ \text{eV} \Rightarrow E_0 = 4.83\ \text{eV} - 0.35\ \text{eV}$$

$E_0 = 4.48$ eV

Q54 **If the photon of wavelength 150 pm strikes an atom, one of its inner bound electrons is ejected out with a velocity of 1.5×10^{7} m s^{-1}. Calculate the energy with which it is bound to the nucleus.**

Solution

$\lambda = 150 \times 10^{-12}\ \text{m} = 1.5 \times 10^{-10}\ \text{m}$; $v = 1.5 \times 10^{7}\ \text{m s}^{-1}$, K.E. $= \frac{1}{2}mv^2$

$$\text{K.E.} = \tfrac{1}{2} \times 9.11 \times 10^{-31}\ \text{kg} \times (1.5 \times 10^{7}\ \text{m s}^{-1})^2 = 4.555 \times 10^{-31} \times 2.25 \times 10^{14}\ \text{kg m}^2\text{s}^{-2}$$

$$\text{K.E.} = 10.25 \times 10^{-17}\ \text{kg m}^2\text{s}^{-2} = 10.25 \times 10^{-17}\ \text{J}$$

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{1.5 \times 10^{-10}\ \text{m}} = 13.252 \times 10^{-16}\ \text{J}$$

$$E = E_0 + \text{K.E.} \Rightarrow E_0 = E - \text{K.E.}$$

$$E_0 = 13.252 \times 10^{-16} - 10.25 \times 10^{-17} = 13.252 \times 10^{-16} - 1.025 \times 10^{-16} = (13.252 - 1.025) \times 10^{-16}$$

$$E_0 = 12.227 \times 10^{-16}\ \text{J} = \frac{12.227 \times 10^{-16}}{1.602 \times 10^{-19}} = 7.63 \times 10^{3}\ \text{eV}$$

$E_0 = 12.227 \times 10^{-16}\ \text{J} = 7.63 \times 10^{3}$ eV

Original notes p. 26

Q55 **Emission transitions in the Paschen series end at orbit n = 3 and start from orbit n and can be represented as $\nu = 3.29 \times 10^{15}\ \text{Hz}\left[\frac{1}{3^2} - \frac{1}{n^2}\right]$. Calculate the value of n if the transition is observed at 1285 nm. Find the region of the spectrum.**

Solution

$$\nu = 3.29 \times 10^{15}\ \text{s}^{-1}\left[\frac{1}{3^2} - \frac{1}{n^2}\right]$$

$$\lambda = 1285 \times 10^{-9}\ \text{m} = 1.285 \times 10^{-6}\ \text{m}, \qquad \nu = \frac{c}{\lambda} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{1.285 \times 10^{-6}\ \text{m}} = 2.3346 \times 10^{14}\ \text{s}^{-1}$$

$$2.3346 \times 10^{14} = 3.29 \times 10^{15}\left[\frac{1}{3^2} - \frac{1}{n^2}\right]$$

$$\frac{2.3346}{32.9} = \frac{1}{3^2} - \frac{1}{n^2} \quad \text{or} \quad 0.071 = \frac{1}{9} - \frac{1}{n^2}$$

$$\frac{1}{n^2} = \frac{1}{9} - 0.071 = 0.111 - 0.071 = 0.04 \qquad n^2 = \frac{1}{0.04} = 25$$

n = 5; Paschen series lies in IR region

Q56 **Calculate the wavelength for the emission transition if it starts from the orbit having radius 1.3225 nm and ends at 211.6 pm. Name the series to which this transition belongs and the region of the spectrum.**

Solution

$$\text{Radius of orbit} = \frac{0.529\ \text{Å} \times n^2}{Z} = \frac{52.9\, n^2}{Z}$$

$$r_1 = 1.3225\ \text{nm} = 1322.5\ \text{pm} = \frac{52.9\, n_1^2}{Z} \text{ — (i)} \qquad r_2 = 211.6\ \text{pm} = \frac{52.9\, n_2^2}{Z} \text{ — (ii)}$$

$$\frac{r_1}{r_2} = \frac{1322.5}{211.6} = \frac{n_1^2}{n_2^2} \qquad \frac{n_1^2}{n_2^2} = 6.25 \quad \text{or} \quad \frac{n_1}{n_2} = (6.25)^{1/2} = 2.5$$

Thus, if $n_1 = 2,\ n_2 = 5$. This transition corresponds to transition from orbit 5 to orbit 2. This means that the transition belongs to the Balmer series. V49

$$\text{Now, } \bar{\nu} = 1.09677 \times 10^7\ \text{m}^{-1} \left[\frac{1}{2^2} - \frac{1}{5^2}\right] = 1.09677 \times 10^7\ \text{m}^{-1} \left[\frac{1}{4} - \frac{1}{25}\right]$$

$$\bar{\nu} = 1.09677 \times 10^7\ \text{m}^{-1} \times \frac{21}{100} = 2.3037 \times 10^6\ \text{m}^{-1}$$

$$\text{Wavelength, } \lambda = \frac{1}{\bar{\nu}} = \frac{1}{2.3037 \times 10^6\ \text{m}^{-1}} = 0.434 \times 10^{-6}\ \text{m} = 434 \times 10^{-9}\ \text{m}$$

$\lambda = 434$ nm; it lies in the visible region (Balmer series)

Q57 **Dual behaviour of matter proposed by de Broglie led to the discovery of electron microscope often used for the highly magnified images of biological molecules and other types of material. If the velocity of the electron in this microscope is $1.6 \times 10^6\ \text{m s}^{-1}$, calculate de Broglie wavelength associated with this electron.**

Solution

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J s}}{9.1 \times 10^{-31}\ \text{kg} \times 1.6 \times 10^6\ \text{m s}^{-1}} = 0.455 \times 10^{-9}\ \text{kg m}^2\text{s}^{-2}\,\text{s m}^{-1}\text{kg}^{-1}\text{s}$$

$\lambda = 0.455$ nm $= 455$ pm

Q58 **Similar to electron diffraction, neutron diffraction microscope is also used for the determination of the structure of molecules. If the wavelength used here is 800 pm, calculate the characteristic velocity associated with the neutron.**

Solution

$\lambda = 800\ \text{pm} = 8 \times 10^{-10}\ \text{m};\ m = 1.675 \times 10^{-27}\ \text{kg}$

$$v = \frac{h}{m\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s}}{1.675 \times 10^{-27}\ \text{kg} \times 8 \times 10^{-10}\ \text{m}}$$

$$v = \frac{6.626 \times 10^{-34+37}}{13.4} = 0.494 \times 10^3 = 494\ \text{m s}^{-1}$$

$v = 494\ \text{m s}^{-1}$

Original notes p. 27

Q59 **If the velocity of the electron in Bohr's first orbit is $2.19 \times 10^6\ \text{m s}^{-1}$, calculate the de Broglie wavelength associated with it.** V50

Solution

$v = 2.19 \times 10^6\ \text{m s}^{-1}$

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J s}}{9.1 \times 10^{-31}\ \text{kg} \times 2.19 \times 10^6\ \text{m s}^{-1}}$$

$\lambda = 0.33243 \times 10^{-9}\ \text{m} = 332.43\ \text{pm}$

Q60 **The velocity associated with a proton moving in a potential difference of 1000 V is $4.37 \times 10^5\ \text{m s}^{-1}$. If a hockey ball of mass 0.1 kg is moving with this velocity, calculate the wavelength associated with this velocity.**

Solution

$v = 4.37 \times 10^5\ \text{m s}^{-1}$; m = 0.1 kg

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J s}}{0.1\ \text{kg} \times 4.37 \times 10^5\ \text{m s}^{-1}}$$

$$\lambda = \frac{6.626}{0.437} \times 10^{-39} = 15.16 \times 10^{-39}\ \text{m}$$

$\lambda = 1.516 \times 10^{-38}\ \text{m}$

Q61 **If the position of the electron is measured within an accuracy of ± 0.002 nm, calculate the uncertainty in the momentum of the electron. Suppose the momentum of the electron is $\frac{h}{4\pi} \times 0.05\ \text{nm}$, is there any problem in defining this value?**

Solution

$\Delta x = 0.002\ \text{nm} = 2 \times 10^{-12}\ \text{m}$

$$\Delta x \times \Delta p = \frac{h}{4\pi} \quad \text{or} \quad \Delta p = \frac{h}{4\pi \times \Delta x}$$

$$\Delta p = \frac{6.626 \times 10^{-34}\ \text{J s}}{4 \times 3.14 \times 2 \times 10^{-12}\ \text{m}} = \frac{6.626 \times 10^{-34} \times 10^{12}\ \text{kg m}^2\text{s}^{-2}\,\text{s}}{25.12\ \text{m}} = 0.263 \times 10^{-22}\ \text{kg m s}^{-1}$$

$$\text{Actual momentum } (p) = \frac{h}{4\pi \times 0.5 \times 10^{-9}\ \text{m}} = \frac{6.626 \times 10^{-34}\ \text{J s}}{4 \times 3.14 \times 0.5 \times 10^{-9}\ \text{m}} = 1.055 \times 10^{-24}\ \text{kg m s}^{-1}$$

V51 *Reproduced as written — please verify the denominator.*

Actual momentum is smaller than uncertainty in measuring momentum. Therefore, momentum of electron cannot be defined.

Q62 **The quantum numbers of six electrons are given below. Arrange them in order of increasing energies. List if any of these combinations has/have same energy.**

Solution

	Quantum numbers	Orbital
(i)	$n = 4,\ l = 2,\ m_l = -2,\ m_s = -\frac{1}{2}$	4d (d_{xy})
(ii)	$n = 3,\ l = 2,\ m_l = 1,\ m_s = +\frac{1}{2}$	3d $(d_{x^2-y^2})$
(iii)	$n = 4,\ l = 1,\ m_l = 0,\ m_s = +\frac{1}{2}$	4p (p_y)
(iv)	$n = 3,\ l = 2,\ m_l = -2,\ m_s = -\frac{1}{2}$	3d (d_{xy})
(v)	$n = 3,\ l = 1,\ m_l = -1,\ m_s = +\frac{1}{2}$	3p (p_y)
(vi)	$n = 4,\ l = 1,\ m_l = 0,\ m_s = +\frac{1}{2}$	4p (p_y)

Increasing order of energies: (v) < (ii) = (iv) < (vi) = (iii) < (i)

Q63 **The bromine atom possesses 35 electrons. It contains 6 electrons in 2p orbital, 6 electrons in 3p orbital and 5 electrons in 4p orbital. Which of these electrons experiences the lowest effective nuclear charge?**

Solution

4p-orbital

Q64 **Among the following pairs of orbitals, which orbital will experience more effective nuclear charge? (i) 2s and 3s (ii) 4d and 4f (iii) 3d and 3p**

Solution

(i) 2s > 3s | (ii) 4d > 4f | (iii) 3p > 3d

Q65 **The unpaired electrons in Al and Si are present in 3p orbital. Which electrons will experience more effective nuclear charge from the nucleus?**

Solution

The unpaired electrons in silicon experience more effective nuclear charge than aluminium because the atomic no. of Si is greater than Al.

Original notes p. 28

Q66 **Indicate the number of unpaired electrons in: (a) P (b) Si (c) Cr (d) Fe (e) Kr**

Solution

Element	Configuration	Unpaired e^-
(a) P (Z = 15)	$[\mathrm{Ne}]\,3s^2\,3p^3$	3
(b) Si (Z = 14)	$[\mathrm{Ne}]\,3s^2\,3p^2$	2
(c) Cr (Z = 24)	$[\mathrm{Ar}]\,4s^1\,3d^5$	6
(d) Fe (Z = 26)	$[\mathrm{Ar}]\,4s^2\,3d^6$	4
(e) Kr (Z = 36)	$[\mathrm{Ar}]\,4s^2\,3d^{10}\,4p^6$	0

Q67 **(a) How many subshells are associated with n = 4? (b) How many electrons will be present in the subshells having m_s value of $-\frac{1}{2}$ for n = 4?**

Solution

(a) For n = 4, No. of subshells: $l = 0,\ l = 1,\ l = 2,\ l = 3$ → **4 subshells**

(b) Total electrons $= 2n^2 = 2(4)^2 = 2 \times 16 = 32\ e^-\mathrm{s}$

(a) 4 subshells | (b) Total electrons with $m_s = -\frac{1}{2}$ is 16

The rest of original page 28 is blank.

Original notes p. 29

2.23 Mq MCQs

Practice MCQs

Set 1

1 **The atomic theory of matter was first proposed by**

(a) John Dalton (b) Rutherford (c) Thomson (d) Bohr

Ans: (a) John Dalton

2 **Positive or canal rays are:**

(i) Electromagnetic waves

(ii) a stream of positively charged gaseous ions

(iii) a stream of electrons

(iv) neutrons

Ans: (ii)

3 **Which of the following is not true for cathode rays?**

(i) Cathode rays consist of negatively charged particles

(ii) Cathode rays were discovered by J.J. Thomson

(iii) Nature of cathode rays depends upon the nature of gas in the discharge tube

(iv) Cathode rays are produced inside the discharge tubes

Ans: (iii)

4 **Which has highest e/m ratio?**

(i) He^{2+} (ii) H^{+} (iii) He^{+} (iv) H

Ans: not filled in the original notes

5 **The charge to the mass ratio of electron was found to be**

(i) $1.6022 \times 10^{-19}\ C\ kg^{-1}$

(ii) $1.925 \times 10^{12}\ C\ kg^{-1}$

(iii) $1.758 \times 10^{11}\ C\ kg^{-1}$

(iv) $1.869 \times 10^{13}\ C\ kg^{-1}$

Ans: (iii)

6 **The ratio of mass of an electron to that of mass of hydrogen atom is**

(i) 1 : 3871 (ii) 1 : 1837 (iii) 1 : 1296 (iv) 1 : 3781

Ans: (ii) 1 : 1837

7 **The discovery of neutron became very late because**

(i) Neutrons are present in the nucleus

(ii) Neutrons are highly unstable particles

(iii) neutrons are chargeless

(iv) neutrons do not move

Ans: (iii)

8 **Identify the incorrect statement among the following**

(i) Electrons are deflected by electric field

(ii) When cathode rays fall on solids such as Cu, X-rays are produced

(iii) Cathode rays produce mechanical effect, as they rotate the wheel placed on their path

(iv) Cathode rays cannot penetrate through thin metal sheets

Ans: (iv)

9 **The ratio of charge and mass would be greater for:**

(i) proton (ii) electron

(iii) neutron (iv) alpha particle

Ans: (ii)

10 **Statement I: e/m value of cathode ray is universal constant while e/m value of positive ray is variable.**
Statement II: The e/m value of positive rays does not depend upon the nature of gas involved in the discharge tube.

(i) Statement I is correct but statement II is incorrect

(ii) Statement I is incorrect and statement II is correct

(iii) Both statements I and II are correct

(iv) Both statements I and II are incorrect

Ans: (i)

V52 *Left-edge characters cut off in the scan were completed.*

Original notes p. 30

Set 2

1 **Discovery of nucleus of an atom was due to experiment carried by**

(i) Bohr (ii) Goldstein (iii) Rutherford (iv) Thomson

Ans: not filled in the original notes

2 **Isotopes have**

(i) Same no. of protons (ii) Same no. of neutrons

(iii) Different no. of electrons (iv) Different atomic numbers

Ans: not filled in the original notes

3 **Species which is isoelectronic with CN^- is**

(i) CO (ii) O^+ (iii) F^- (iv) O_2 V53

Ans: not filled in the original notes

4 **Identify the pair of molecules which is isoelectronic:**

(i) N_2 and CO (ii) N_2 and CS (iii) O_2 and CO (iv) O_2 and CS

Ans: not filled in the original notes

5 **When a gold sheet is bombarded by a beam of α-particles, only a few of them get deflected whereas most go straight, undeflected. This is because,**

(i) The force of attraction exerted on α-particles by the oppositely charged electrons is not sufficient.

(ii) A nucleus has a much smaller volume than that of atom.

(iii) The force of repulsion acting on the fast moving α-particles is very small.

(iv) The neutrons in the nucleus do not have any effect on the α-particles.

Ans: not filled in the original notes

6 **The incorrect statements regarding the Thomson's model of atom are:**

(i) in an atom, the mass of the atom is uniformly distributed.

(ii) the positive charge is uniformly distributed.

(iii) in the positively charged sphere the electrons are unevenly distributed.

(iv) The electrons attract each other to stabilise the atom.

Ans: not filled in the original notes

7 **Rutherford's experiment on scattering of alpha particles showed for the first time that atom has**

(i) Nucleus **(ii)** Protons **(iii)** Electrons **(iv)** Neutrons

Ans: not filled in the original notes

8 **Match List I with List II and choose the correct option:**

List I		List II	
(A)	$^{76}_{32}Ge$, $^{77}_{33}As$	(I)	Isotopes
(B)	$^{57}_{26}Fe$, $^{58}_{26}Fe$	(II)	Monoelectronic
(C)	H, H^+	(III)	Isobars
(D)	$^{3}_{1}H$, $^{3}_{2}He$	(IV)	Isotones

(i) A-IV, B-I, C-II, D-III **(ii)** A-II, B-I, C-IV, D-...

(iii) A-IV, B-III, C-II, D-I **(iv)** A-II, B-III, C-IV, D-...

Ans: not filled in the original notes V54

9 **Statement I: Isotopes have same chemical properties but different physical properties.**
Statement II: Isobars have different chemical properties but few common physical properties.

(i) Statement I is correct but statement II is incorrect

(ii) Statement I is incorrect but statement II is correct

(iii) Both statements I and II are correct

(iv) Both statements I and II are incorrect

Ans: not filled in the original notes

10 **Assertion (A):** $^{13}_{6}C$, $^{14}_{7}N$ **and** $^{4}_{2}He$, $^{5}_{3}Li$ **are pairs of isotones.** V55
Reason (R): Isotones have different no. of neutrons.

(i) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of A

(ii) Both A and R are true but R is not a correct explanation of A

(iii) Assertion is true and R is false

(iv) Assertion is false and Reason is true

Ans: not filled in the original notes

Original notes p. 31

Set 3

1 **Which of the following has maximum wavelength?**

(i) Cosmic rays (ii) X-rays (iii) Micro waves (iv) Radio waves

Ans: not filled in the original notes

2 **The correct order of increasing frequency of electromagnetic radiations is**

(i) Microwave < UV < IR < X-rays

(ii) Microwave < Radiowave < Visible < X-rays

(iii) UV < Radiowave < X-ray < γ-rays

(iv) Radiowave < IR < Visible < X-rays

Ans: not filled in the original notes

3 **The quantum of light is called**

(i) Photon (ii) Neutron (iii) Electron (iv) Proton

Ans: not filled in the original notes

4 **The energy of photon of light is ____ proportional to its frequency and ____ proportional to its wavelength.**

(i) directly, directly (ii) inversely, inversely

(iii) inversely, directly (iv) directly, inversely

Ans: not filled in the original notes

5 **The no. of photons of light of wavelength 7000 Å equivalent to 1 J are**

(i) 3.52×10^{-18} (ii) 3.52×10^{18} (iii) 50000 (iv) 10,000

Ans: not filled in the original notes

6 **A 100 watt bulb emits monochromatic light of wavelength 400 nm. The no. of photons emitted per second by the bulb is**

(i) 2.022×10^{23} (ii) 2.012×10^{20} (iii) 20.21×10^{21} (iv) 22.06×10^{20}

Ans: not filled in the original notes

7 **What is the wavelength of a photon with energy 3×10^{-19} J, according to Planck's quantum theory?**

(i) 1.15×10^{-7} m (ii) 6.63×10^{-7} m

(iii) 6.63×10^{-8} m (iv) 9.23×10^{-7} m

Ans: not filled in the original notes

8 **Identify the incorrect statement among the following**

(i) Electromagnetic radiations do not require any medium for their propagation

(ii) In electromagnetic radiations electric and magnetic fields are $\perp$ to each other

(iii) The height of crest or depth of trough of a wave is called amplitude

(iv) The number of waves passing through a given point in one second is called wave number

Ans: not filled in the original notes

9 **Statement I: As the wavelength is doubled, energy of radiation is also doubled.**
Statement II: $E = \dfrac{h}{\lambda c}$ is energy of one quantum.

Options and answer not written in the original notes

10 **Assertion (A): A body that is capable of absorbing all types of radiations is called a black body.**
Reason (R): When a black body is heated, it emits thermal radiations of different wavelengths or frequencies.

Options and answer not written in the original notes

V56 *Left-edge characters cut off in the scan were completed.*

Original notes p. 32

2.24 PYQ NEET PYQ

Structure of Atom — NEET PYQs

1 **Select the correct statement from the following:**
(A) Atoms of all elements are composed of two fundamental particles.
(B) The mass of electron is 9.10939×10^{-31} kg.
(C) All the isotopes of a given element show same chemical properties.
(D) Protons and electrons are collectively known as nucleons.
(E) Dalton's atomic theory regarded the atom as an ultimate particle of matter.
Choose the correct option:

(a) A and E only
(b) B, C and E only
(c) A, B and C only
(d) C, D and E only

Ans: (b) B, C and E

2 **From the following pairs of ions which one is not an iso-electronic pair?**

(a) Fe^{2+}, Mn^{2+}
(b) O^{2-}, F^{-}
(c) Na^{+}, Mg^{2+}
(d) Mn^{2+}, Fe^{3+}

Ans: (a) Fe^{2+}, Mn^{2+}

3 **The number of protons, neutrons and electrons in $^{175}_{71}Lu$ respectively are**

(a) 71, 104 and 71
(b) 104, 71 and 71
(c) 71, 71 and 104
(d) 175, 104 and 71

Ans: (a) 71, 104, 71

4 Be^{2+} **is isoelectronic with which of the following ions?**

(a) H^{+}
(b) Li^{+}
(c) Na^{+}
(d) Mg^{2+}

Ans: (b) Li^{+}

5 **Isoelectronic species are**

(a) CO, CN^{-}, NO^{+}, C_2^{2-}
(b) CO^{-}, CN, NO, C_2^{-}
(c) CO^{+}, CN^{+}, NO^{-}, C_2
(d) CO, CN, NO, C_2

Ans: (a)

6 **The ion that is isoelectronic with CO is** V57

(a) CN^- (b) N_2^+ (c) O^{2-} (d) N_2^-

Ans: (a) CN^-

7 **Which one of the following is not isoelectronic with** O^{2-}**?**

(a) Ti^+ (b) Na^+ (c) N^{3-} (d) F^-

Ans: (a)

8 **The ratio of the wavelength of the light absorbed by a hydrogen atom when it undergoes n = 2 → n = 3 and n = 4 → n = 6 transition, respectively, is**

(a) $\frac{1}{9}$ (b) $\frac{1}{4}$ (c) $\frac{1}{36}$ (d) $\frac{1}{16}$

Ans: (b) $\frac{1}{4}$

Solution

For n = 2 to n = 3:

$$\frac{1}{\lambda_1} = R_H\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = R_H\left[\frac{9-4}{4\times 9}\right] = R_H\left[\frac{5}{36}\right] \quad -(\text{i})$$

$$\frac{1}{\lambda_2} = R_H\left[\frac{1}{4^2} - \frac{1}{6^2}\right] = R_H\left[\frac{36-16}{36\times 16}\right] = R_H\left[\frac{20}{576}\right]$$

$$\frac{1}{\lambda_2} = R_H\left[\frac{5}{144}\right] \quad -(\text{ii})$$

On dividing eq. (ii) by (i):

$$\frac{\lambda_1}{\lambda_2} = \frac{R_H\left[\frac{5}{144}\right]}{R_H\left[\frac{5}{36}\right]} = \frac{5}{144}\times\frac{36}{5} = \frac{36}{144}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{1}{4}$$

V58 *Line endings cut off in the scan were completed.*

9 **A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1368 kHz. The wavelength of the electromagnetic radiations emitted by the transmitter is**

(a) 21.92 cm (b) 219.3 m (c) 219.2 m (d) 21... V59

Ans: (b) 219.3 m

Solution

$$\lambda = \frac{c}{\nu} = \frac{3\times 10^8\ \text{m s}^{-1}}{1368\times 10^3\ \text{s}^{-1}} = 0.0021929\times 10^5\ \text{m} = 219.29 = 219.3\ \text{m}$$

Original notes p. 33

10 **Given below are two statements:**
Statement I: The Balmer spectral line for H-atom with lowest energy is located at $\frac{5}{36}R_H$ cm^{-1}.
Statement II: When the temperature of black body increases, the maxima of the curve (intensity vs wavelength) shifts to shorter wavelength.
Choose the correct answer from the options given below: V60

(a) Statement I is true but II is false

(b) Statement I is false and II is correct

(c) Both statements I and II are correct

(d) Both statements I and II are incorrect

Ans: (c)

11 **Which of the following series of transitions in the spectrum of hydrogen atom falls in the visible region?**

(a) Brackett series (b) Lyman series

(c) Balmer series (d) Paschen series

Ans: (c) Balmer series

12 **Calculate the energy in joule corresponding to light of wavelength 45 nm.**

(a) 6.67×10^{15} (b) 6.67×10^{11} (c) 4.42×10^{-15} (d) 4.42×10^{-18}

Ans: (d) 4.42×10^{-18}

Solution

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}\ \text{m s}^{-1}}{45 \times 10^{-9}\ \text{m}}$$

$$E = \frac{19.878}{45} \times 10^{-17} = 0.442 \times 10^{-17} = 4.42 \times 10^{-18}\ \text{J}$$

13 **The value of Planck's constant is 6.63×10^{-34} J s. The speed of light is 3×10^{17} nm s^{-1}. Which value is closest to the wavelength in nanometre of a quantum of light with frequency of 6×10^{15} s^{-1}?**

(a) 50 (b) 75 (c) 10 (d) 25

Ans: (a) 50

Solution

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^{17}\ \text{nm s}^{-1}}{6 \times 10^{15}\ \text{s}^{-1}} = 0.5 \times 10^{2}\ \text{nm} = 50\ \text{nm}$$

14 **According to law of photochemical equivalence the energy absorbed (in erg/mole) is given as** $(h = 6.626 \times 10^{-27}\ \text{erg s},\ c = 3 \times 10^{10}\ \text{cm s}^{-1},\ N_A = 6.022 \times 10^{23}\ \text{mol}^{-1})$

(a) $\frac{1.196 \times 10^{8}}{\lambda}$ (b) $\frac{2.859 \times 10^{5}}{\lambda}$ (c) $\frac{2.859 \times 10^{16}}{\lambda}$ (d) $\frac{1.196 \times 10^{16}}{\lambda}$

Ans: (a) $\frac{1.196 \times 10^{8}}{\lambda}$

Solution

$$E = \frac{N_A hc}{\lambda} = \frac{6.022 \times 10^{23} \times 3 \times 10^{10} \times 6.626 \times 10^{-27}}{\lambda}$$

$$E = \frac{119.7 \times 10^{6}\ \text{erg s mol}^{-1}}{\lambda} = \frac{1.197 \times 10^{8}}{\lambda}$$

15 **The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relationship between their wavelengths i.e. λ_1 and λ_2 will be**

(a) $\lambda_1 = \lambda_2$ (b) $\lambda_1 = 2\lambda_2$ (c) $\lambda_1 = 4\lambda_2$ (d) $\lambda_1 = \frac{1}{2}\lambda_2$

Ans: (b) $\lambda_1 = 2\lambda_2$

Solution

$$E_1 = \frac{hc}{\lambda_1} \text{ and } E_2 = \frac{hc}{\lambda_2} \qquad \frac{E_1}{E_2} = \frac{hc}{\lambda_1} \times \frac{\lambda_2}{hc} = \frac{\lambda_2}{\lambda_1}$$

$$\frac{25}{50} = \frac{\lambda_2}{\lambda_1} \Rightarrow \frac{1}{2} = \frac{\lambda_2}{\lambda_1} \Rightarrow \lambda_1 = 2\lambda_2$$

16 **The value of Planck's constant is 6.63×10^{-34} J s. The velocity of light is 3.0×10^{8} m s^{-1}. Which value is closest to the wavelength in nanometre of a quantum of light with frequency 8×10^{15} s^{-1}?**

(a) 2×10^{-25} (b) 5×10^{-18} (c) 4×10^{1} (d) 3×10^{7}

Ans: (c) 4×10^{1}

Solution

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{8 \times 10^{15}\ \text{s}^{-1}} = 0.375 \times 10^{-7} = 3.75 \times 10^{-8} = 37.5 \times 10^{-9}\ \text{m near to } 4 \times 10\ \text{nm}$$

Original notes p. 34

V61 *This original page is faint — values read carefully; please verify.*

17 **For given energy, $E = 3.03 \times 10^{-19}$ J, corresponding wavelength is**

(a) 65.6 nm (b) 6.56 nm (c) 3.4 nm (d) 656 nm

Ans: (d) 656 nm

Solution

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{6.626 \times 10^{-34}\ \text{J s} \times 3 \times 10^{8}}{3.03 \times 10^{-19}\ \text{J}}$$

$$\lambda = 6.56 \times 10^{-7}\ \text{m} = 656\ \text{nm}$$

18 **What will be the longest wavelength line in Balmer series of spectrum?**

(a) 546 nm **(b)** 656 nm **(c)** 566 nm **(d)** 556 nm

Ans: (b) 656 nm

Solution

Longest λ, minimum E: $n_1 = 2,\ n_2 = 3$

$$\frac{1}{\lambda} = 109677\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = 109677\left[\frac{9-4}{36}\right] = 109677 \times \frac{5}{36}$$

$$\frac{1}{\lambda} = 15233 \Rightarrow \lambda = \frac{1}{15233} = 6.56 \times 10^{-5}\ \text{cm} = 656\ \text{nm}$$

19 **Energy and radius of first Bohr orbit of He^+ and Li^{2+} are [Given $R_H = 2.18 \times 10^{-18}$ J, $a_0 = 52.9$ pm]**

(a) $E_n(Li^{2+}) = -19.62 \times 10^{-16}$ J; $r_n(Li^{2+}) = 17.6$ pm; $E_n(He^+) = -8.72 \times 10^{-16}$ J; $r_n(He^+) = 26.4$ pm

(b) $E_n(Li^{2+}) = -8.72 \times 10^{-16}$ J; $r_n(Li^{2+}) = 17.6$ pm; $E_n(He^+) = -19.62 \times 10^{-16}$ J; $r_n(He^+) = 17.6$ pm

(c) $E_n(Li^{2+}) = -19.62 \times 10^{-18}$ J; $r_n(Li^{2+}) = 17.6$ pm; $E_n(He^+) = -8.72 \times 10^{-18}$ J; $r_n(He^+) = 26.4$ pm

(d) $E_n(Li^{2+}) = -8.72 \times 10^{-18}$ J; $r_n(Li^{2+}) = 26.4$ pm; $E_n(He^+) = -19.62 \times 10^{-18}$ J; $r_n(He^+) = 17.6$ pm

Ans: (c) $E_n(Li^{2+}) = -19.62 \times 10^{-18}$ J, $r_n(Li^{2+}) = 17.6$ pm; $E_n(He^+) = -8.72 \times 10^{-18}$ J, $r_n(He^+) = 26.4$ pm

Solution

$$E_n(Li^{2+}) = -2.18 \times 10^{-18} \times 9 = -19.62 \times 10^{-18}\ \text{J} \qquad r_n(Li^{2+}) = 52.9 \times \tfrac{1}{3}\ \text{pm} = 17.63\ \text{pm}$$

$$E_n(He^+) = -2.18 \times 10^{-18} \times 4 = -8.72 \times 10^{-18}\ \text{J} \qquad r_n(He^+) = 52.9 \times \tfrac{1}{2} = 26.45\ \text{pm}$$

20 **The energy of an electron in the ground state (n = 1) for He^+ ion is –x J, then that for an electron in n = 2 state for Be^{3+} ion in J is**

(a) –x **(b)** $-\frac{x}{9}$ **(c)** –4x **(d)** $-\frac{4}{9}x$

Ans: (a) –x

Solution

$$E_n = -R_H \frac{Z^2}{n^2}$$

For ground state: Z = 2, n = 1: $E_1 = -R_H \dfrac{2^2}{1^2} = -4R_H = -x$

For excited state: Z = 4, n = 2: $E_2 = -R_H \dfrac{4^2}{2^2} = -R_H \dfrac{16}{4} = -R_H\, 4$

$E_2 = -x$ J

21 **If radius of second Bohr orbit of He^+ ion is 105.8 pm, what is the radius of third Bohr orbit of Li^{2+} ion?**

(a) 158.7 pm **(b)** 15.87 pm **(c)** 1.587 pm **(d)** 158.7 Å

Ans: (a) 158.7 pm

Solution

Radius $= r_0 \times \frac{n^2}{Z}$

For He^+: n = 2, Z = 2; Radius $= r_0 \times \frac{2^2}{2} = \frac{4r_0}{2} = 2r_0$; $r_0 = \frac{\text{Radius}}{2} = \frac{105.8}{2}$ pm

For Li^{2+} ion: n = 3, Z = 3; Radius $= r_0 \times \frac{3^2}{3} = \frac{9r_0}{3} = 3r_0$

$$\text{Radius} = 3 \times \frac{105.8}{2} = 158.7 \text{ pm} = 1.587 \text{ Å}$$

V62 *Last value partly cut off in the scan.*

The lower part of the right column of original page 34 is blank.

Original notes p. 35

22 **Based on equation $E = -2.178 \times 10^{-18}$ J $\left[\frac{Z^2}{n^2}\right]$, certain conclusions are written. Which of them is not correct?** V63

(a) Equation can be used to calculate the change in energy when the electron changes orbit.

(b) For n = 1, the electron has a more negative energy than it does for n = 6 which means that the electron is more loosely bound in the smallest allowed orbit.

(c) The negative sign in equation simply means that the electron's energy bound to the nucleus is lower than it would be if the electron were at the infinite distance from the nucleus.

(d) Larger the value of n, the larger is the orbit radius.

Ans: (b)

23 **According to Bohr's theory which of the following transitions in the hydrogen atom will give rise to the least energetic photon?**

(a) n = 6 to n = 1 **(b)** n = 5 to n = 4

(c) n = 6 to n = 5 **(d)** n = 5 to n = 3

Ans: (c) n = 6 to n = 5

24 **The energy of second Bohr orbit of the hydrogen atom is $-328\ \text{kJ mol}^{-1}$, hence the energy of fourth Bohr orbit would be**

(a) $-41\ \text{kJ mol}^{-1}$ **(b)** $-82\ \text{kJ mol}^{-1}$

(c) $-164\ \text{kJ mol}^{-1}$ **(d)** $-1312\ \text{kJ mol}^{-1}$

Ans: (b) $-82\ \text{kJ mol}^{-1}$

Solution

$$E_n \propto \frac{1}{n^2} \qquad \therefore \frac{E_{n_1}}{E_{n_2}} = \frac{n_2^2}{n_1^2} \qquad E_{n_2} = E_{n_1}\left(\frac{n_1^2}{n_2^2}\right) = E_{n_1}\left(\frac{n_1}{n_2}\right)^2$$

$$E_4 = -328 \times \frac{2^2}{4^2} = -328 \times \frac{4}{16} = -82\ \text{kJ mol}^{-1}$$

25 **The frequency of radiations emitted when electron falls from n = 4 to n = 1 in a hydrogen atom will be (Given ionisation enthalpy of H $= 2.18 \times 10^{-18}\ \text{J atom}^{-1}$ and $h = 6.626 \times 10^{-34}\ \text{J s}$)**

(a) $1.54 \times 10^{15}\ \text{s}^{-1}$ **(b)** $1.03 \times 10^{15}\ \text{s}^{-1}$

(c) $3.08 \times 10^{15}\ \text{s}^{-1}$ **(d)** $2.00 \times 10^{15}\ \text{s}^{-1}$

Ans: (c) $3.08 \times 10^{15}\ \text{s}^{-1}$

Solution

$E = h\nu$ or $\nu = E/h$. For H-atom, $E = -\dfrac{2.18 \times 10^{-18}}{n^2}\ \text{J atom}^{-1}$

$$\Delta E = -2.18 \times 10^{-18}\left[\frac{1}{n_2^2} - \frac{1}{n_1^2}\right]$$

$$\Delta E = -2.18 \times 10^{-18}\left[\frac{1}{4^2} - \frac{1}{1^2}\right] = 2.18 \times 10^{-18}\left[\frac{1}{16} - \ldots\right]$$

$$\Delta E = -2.18 \times 10^{-18}\left[\frac{-15}{16}\right] \qquad \Delta E = -2.18 \times 10^{-18} \times \frac{-15}{16} = 2.044 \times 10^{-18}$$

$$\Delta E = 20.44 \times 10^{-19}\ \text{J atom}^{-1} \qquad \nu = \frac{\Delta E}{h} = \frac{20.44 \times 10^{-19}\ \text{J}}{6.626 \times 10^{-34}\ \text{J s}} = 3.08 \times 10^{15}\ \text{s}^{-1}$$

V64 *Part of one line is cut off in the scan.*

26 **In hydrogen atom, energy of first excited state is –3.4 eV. Then find out K.E. of same orbit of hydrogen atom.**

(a) +3.4 eV **(b)** +6.8 eV **(c)** –13.6 eV **(d)** +13.6 eV

Ans: (a) 3.4 eV

Solution

K.E. $= \frac{1}{2}mv^2$. In Bohr model, there is a specific relationship between Total Energy, K.E. and Potential Energy (P.E.):

① Total energy = K.E. + P.E. ② Relation: K.E. is always equal to –ve total E

$$\text{K.E.} = -E = -(-3.4\ \text{eV}) = +3.4\ \text{eV}$$

Original notes p. 36

27 **Who modified Bohr's theory by introducing elliptical orbits for electron path?**

(a) Rutherford (b) Thomson (c) Hund (d) Sommerfeld

Ans: (d) Sommerfeld V65

28 **The Bohr orbit radius for the hydrogen atom (n = 1) is approximately 0.530 Å. The radius for the excited state (n = 2) orbit is (in Å)**

(a) 4.77 (b) 1.06 (c) 0.13 (d) 2.12

Ans: (d) 2.12

Solution

For nth orbit of H atom, $r_n = n^2 \times r_1$. For 2nd Bohr orbit, $r_n = (2)^2 \times r_1 = 4r_1 = 4 \times 0.530\ \text{Å} = 2.12\ \text{Å}$

29 **In Bohr's model of an atom, when an electron jumps from n = 1 to n = 3, how much energy will be emitted or absorbed?**

(a) 2.389×10^{-12} ergs (b) 0.239×10^{-10} ergs

(c) 2.15×10^{-11} ergs (d) 0.1936×10^{-10} ergs

Ans: (c) 0.1936×10^{-10} **ergs** V66

Solution

$$E_n = -\frac{2.18 \times 10^{-11}}{n^2}\ \text{ergs} \qquad \Delta E = E_3 - E_1 = -2.18 \times 10^{-11}\left[\frac{1}{3^2} - \frac{1}{1^2}\right]$$

$$\Delta E = -2.18 \times 10^{-11}\left[\frac{1-9}{9}\right] = 2.18 \times 10^{-11} \times \frac{8}{9} \qquad \Delta E = 1.937 \times 10^{-11}\ \text{ergs}$$

30 **The radius of hydrogen atom in the ground state is 0.53 Å. The radius of Li^{2+} ion (at. no. = 3) in a similar state is**

(a) 0.53 Å (b) 1.06 Å (c) 0.17 Å (d) 0.2... V67

Ans: (c) 0.17 Å

Solution

$$r_H = 0.53\ \text{Å} \qquad \text{radius of } Li^{2+} = r \times \frac{n^2}{Z} = \frac{0.53 \times (1)^2}{3} = 0.17\ \text{Å}$$

31 **The energy of an electron in the nth Bohr orbit of hydrogen atom is**

(a) $\frac{13.6}{n^4}$ eV (b) $\frac{13.6}{n^3}$ eV (c) $\frac{13.6}{n^2}$ eV (d) $\frac{13.6}{n}$ eV

Ans: (c) $\frac{13.6}{n^2}$ **eV**

32 **The spectrum of He is expected to be similar to that of**

(a) H (b) Li^+ (c) Na (d) He^+

Ans: (b) Li^+

33 **If r is the radius of the first orbit, the radius of nth orbit of H atom is given by**

(a) rn^2 (b) rn (c) r/n (d) r^2n^2

Ans: (a) rn^2

34 **In a hydrogen atom, the de Broglie wavelength of an electron in the second Bohr orbit is**

(a) 211.6 pm (b) 211.6π pm (c) 52.9π pm (d) 105.8 pm

Ans: (b) 211.6π pm

Solution

$$r_2 = 52.9 \times 2^2 = 211.6 \text{ pm} \qquad 2\lambda = 2\pi r_2 \Rightarrow \lambda = \pi r_2 \Rightarrow \lambda = \pi \times 211.6 \text{ pm}$$

Original notes p. 37

35 **A 0.66 kg ball is moving with a speed of 100 m s^{-1}. The associated wavelength is** $(h = 6.6 \times 10^{-34}$ J s) V68

(a) 6.6×10^{-32} m (b) 6.6×10^{-34} m (c) 1×10^{-35} m (d) 1×10^{-32} m

Ans: (c) 1×10^{-35} m

Solution

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J s}}{0.66 \times 100} = 1 \times 10^{-35} \text{ m}$$

36 **If uncertainty in position and momentum are equal, then uncertainty in velocity is**

(a) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$ (b) $\sqrt{\frac{h}{\pi}}$ (c) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (d) $\sqrt{\frac{h}{\dots}}$ V69

Ans: (c) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

Solution

$$\Delta x = \Delta p \qquad \Delta x \cdot \Delta p = \frac{h}{4\pi} \Rightarrow \Delta p \cdot \Delta p = \frac{h}{4\pi}$$

$$(\Delta p)^2 = \frac{h}{4\pi} \Rightarrow \Delta p = \sqrt{\frac{h}{4\pi}} = \frac{1}{2}\sqrt{\frac{h}{\pi}}$$

$$\text{But } \Delta p = m \times \Delta v \qquad m \times \Delta v = \frac{1}{2}\sqrt{\frac{h}{\pi}} \qquad \Delta v = \frac{1}{2m}\sqrt{\frac{h}{\pi}}$$

37 **The measurement of the electron position is associated with an uncertainty in momentum which is equal to 1×10^{-18} g cm s^{-1}. The uncertainty in electron velocity is $(m = 9 \times 10^{-28}$ g)**

(a) 1×10^{5} cm s^{-1} **(b)** 1×10^{11} cm s^{-1}

(c) 1×10^{9} cm s^{-1} **(d)** 1×10^{6} cm s^{-1}

Ans: (c) 1×10^{9} cm s^{-1}

Solution

$$\Delta p = m \times \Delta v = 1 \times 10^{-18} \text{ g cm s}^{-1} \qquad \Delta v = \frac{1 \times 10^{-18} \text{ g cm s}^{-1}}{9 \times 10^{-28} \text{ g}} = 1.1 \times 10^{9} \text{ cm s}^{-1}$$

38 **The mass of electron is 9.11×10^{-31} kg. Planck constant is 6.626×10^{-34} J s; the uncertainty involved in the measurement of velocity within the distance of 0.1 Å is**

(a) 5.79×10^{5} m s^{-1} **(b)** 5.79×10^{6} m s^{-1}

(c) 5.79×10^{7} m s^{-1} **(d)** 5.79×10^{8} m s^{-1}

Ans: (b) 5.79×10^{6} m s^{-1}

Solution

$$\Delta x \times m\Delta v = \frac{h}{4\pi} \qquad 0.1 \times 10^{-10} \text{ m} \times 9.11 \times 10^{-31} \text{ kg } \Delta v = \frac{h}{4 \times 3.14}$$

$$\Delta v = \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 0.1 \times 10^{-10} \times 9.11 \times 10^{-31}} = 5.79 \times 10^{6} \text{ m s}^{-1}$$

39 **The uncertainty in momentum of an electron is 1×10^{-5} kg m/s. The uncertainty in its position will be $(h = 6.62 \times 10^{-34}$ kg m^2/s)**

(a) 5.27×10^{-30} m **(b)** 1.05×10^{-26} m

(c) 1.05×10^{-28} m **(d)** 5.25×10^{-28} m

Ans: (a) 5.27×10^{-30} m

Solution

$$\Delta x = \frac{h}{4\pi \times \Delta p} = \frac{6.626 \times 10^{-34} \text{ J s}}{4 \times 3.14 \times 1 \times 10^{-5}} \qquad \Delta x = 5.27 \times 10^{-28} \text{ m}$$

V70 *Exponent of the last line reproduced as read — please verify.*

40 **The de Broglie wavelength of a particle with mass 1 g and velocity 100 m/s is**

(a) 6.63×10^{-35} m (b) 6.63×10^{-34} m

(c) 6.63×10^{-33} m (d) 6.65×10^{-35} m

Ans: (c) 6.63×10^{-33} m

Solution

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-27} \text{ ergs}}{1 \text{ g} \times 10000 \text{ cm s}^{-1}} \qquad \lambda = 6.63 \times 10^{-31} \text{ cm} = 6.63 \times 10^{-33} \text{ m}$$

Original notes p. 38

41 **The position of both an electron and a helium atom is known within 1.0 nm. Further the momentum of the electron is known within 5×10^{-26} kg m s^{-1}. The minimum uncertainty in the measurement of the momentum of the helium atom is**

(a) 8×10^{-26} kg m s^{-1} (b) 80 kg m s^{-1}

(c) 50 kg m s^{-1} (d) 5.0×10^{-26} kg m s^{-1}

Ans: (d) 5.0×10^{-26} kg m s^{-1}

Solution

$\Delta x \times \Delta p = \frac{h}{4\pi}$. As Δx = 1.0 nm for e^- and He atom, so Δp is also same.

42 **Uncertainty in position of an electron (mass $= 9.1 \times 10^{-28}$ g) moving with a velocity of 3×10^4 cm/s accurate up to 0.001% will be (Use $h/4\pi$ in uncertainty expression where $h = 6.626 \times 10^{-27}$ ergs)**

(a) 5.76 cm (b) 7.68 cm (c) 1.93 cm (d) 3.84 cm

Ans: (c) 1.93 cm

Solution

Mass of $e^- = 9.1 \times 10^{-28}$ g; Velocity of $e^- = 3 \times 10^4$ cm s^{-1}; Accuracy = 0.001% and $h = 6.626 \times 10^{-34}$ ergs

$$\text{Actual velocity} = 3 \times 10^4 \text{ cm s}^{-1} \times \frac{0.001}{100} \qquad \Delta v = 3 \times 10^4 \times 10^{-5} = 0.3 \text{ cm/s}$$

$$\therefore \Delta x = \frac{h}{4\pi m \Delta v} = \frac{6.626 \times 10^{-34} \text{ J s}}{4 \times 3.14 \times 9.1 \times 10^{-28} \times 0.3} \qquad \Delta x = 1.93 \text{ cm}$$

V71 *Value of h in the solution reproduced as written — please verify.*

43 **Which of the following statements does not form a part of Bohr's model of hydrogen atom?**

(a) Energy of the electrons in orbits are quantised.

(b) The electron in the orbit nearest the nucleus has the lowest energy.

(c) Electrons revolve in different orbits around the nucleus.

(d) The position and velocity of the electron in the orbit cannot be determined simultaneously.

Ans: (d)

44 **The quantum numbers of four electrons are given below:**

(I) $n = 4,\ l = 2,\ m_l = -2,\ m_s = -\frac{1}{2}$ **(II)** $n = 3,\ l = 2,\ m_l = 1,\ m_s = +\frac{1}{2}$

(III) $n = 4,\ l = 1,\ m_l = 0,\ m_s = +\frac{1}{2}$ **(IV)** $n = 3,\ l = 1,\ m_l = -1,\ s = +\frac{1}{2}$

The correct decreasing order of the energy of these electrons is

(a) IV > II > III > I

(b) I > III > II > IV

(c) III > I > II > IV

(d) I > II > III > IV

Ans: (b) I > III > II > IV

Solution

I = 4d, II = 3d, III = 4p, IV = 3p ⇒ 4d > 4p > 3d > 3p

45 **Match List I with List II:**

List I		List II	
A	m_l	I	Shape of orbital
B	m_s	II	Size of orbital
C	l	III	Orientation of orbital
D	n	IV	Orientation of spin of electron

(a) A-I, B-III, C-II, D-IV

(b) A-III, B-IV, C-I, D-II

(c) A-III, B-IV, C-II, D-I

(d) A-II, B-I, C-IV, D-III

Ans: (b)

Original notes p. 39

46 **[Top of question cut off in the scan] ... set of quantum no. from the following:** V72

(a) $n = 4,\ l = 3,\ m_l = -3, -2, -1, 0, +1, +2, +3,\ m_s = -\frac{1}{2}$

(b) $n = 5,\ l = 2,\ m_l = -2, -1, +1, +2,\ m_s = +\frac{1}{2}$

(c) $n = 4,\ l = 2,\ m_l = -2, -1, 0, +1, +2,\ m_s = -\frac{1}{2}$

(d) $n = 5,\ l = 3,\ m_l = -3, -2, -1, 0, +1, +2, +3,\ m_s = +\frac{1}{2}$

Ans: (b)

Solution

$m_l = (2l + 1)$ for $l = 2$: value $= 2 \times 2 + 1 = 5$

47 **Given below are two statements:**
Statement I: The value of wave function ψ depends upon the co-ordinates of the electron in the atom.
Statement II: The probability of finding an electron at a point within an atom is proportional to the orbital wave function.

(a) Statement I is correct but II is incorrect

(b) Statement I is false and II is true

(c) Both statements I and II are true

(d) Both statements I and II are false

Ans: (a)

48 **The relation between n_m (no. of permissible values of magnetic Q. No.) for a given value of azimuthal Q. No. is**

(a) $n_m = 2l^2 + 1$ **(b)** $n_m = l + 2$ **(c)** $l = \frac{n_m - 1}{2}$ **(d)** $l = 2n_m + 1$

Ans: (c)

49 **The orbital having 3 angular nodes and 3 total nodes is**

(a) 5p **(b)** 3d **(c)** 4f **(d)** 6d

Ans: (c) 4f

Solution

No. of spherical/radial nodes in any orbital $= n - l - 1$

No. of planar/angular nodes in orbital $= l$

50 **Identify the incorrect statement from the following:**

(a) All the five 5d orbitals are different in size when compared to the respective 4d orbitals.

(b) All the five 4d orbitals have shapes similar to the respective 3d orbitals.

(c) In an atom, all the five 3d orbitals are equal in energy in free state.

(d) The shapes of d_{xy}, d_{yz}, d_{zx} orbitals are similar to each other and $d_{x^2-y^2}$ and d_{z^2} are similar to each other.

Ans: (d)

51 **4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The correct option is**

(a) 5f > 6p > 4d > 5p

(b) 5f > 6p > 5p > 4d

(c) 6p > 5f > 5p > 4d

(d) 6p > 5f > 4d > 5p

Ans: (b)

52 **Which of the following is wrong?**

(a) Total orbital angular momentum of electron in s-orbital is equal to zero.

(b) An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.

(c) The electronic configuration of N atom is $1s^2\ 2s^2\ 2p^3$:

(d) The value of m for d_{z^2} is zero.

Ans: (c)

53 **How many electrons can fit in the orbital for which n = 3, l = 1?**

(a) 2

(b) 6

(c) 10

(d) 14

Ans: 2

Original notes p. 40

54 **Which one is the wrong statement?**

(a) The uncertainty principle is $\Delta E \times \Delta t \geq \frac{h}{4\pi}$

(b) Half filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.

(c) The energy of 2s-orbital is less than the energy of 2p-orbital in case of hydrogen like atoms.

(d) de Broglie's wavelength is given by $\lambda = \frac{h}{mv}$

Ans: (c) Energy depends upon value of n

55 **Which of the following pairs of d-orbitals will have electron density along the axes?**

(a) d_{z^2}, d_{xz} (b) d_{xz}, d_{yz} (c) d_{z^2}, $d_{x^2-y^2}$ (d) d_{xy}, $d_{x^2-y^2}$

Ans: (c)

56 **Two electrons occupying the same orbital are distinguished by**

(a) Azimuthal Quantum No. (b) Spin Quantum No.

(c) Principal Quantum No. (d) Magnetic Q. No.

Ans: (b)

57 **Which is the correct order of increasing energy of the listed orbitals in the atom of titanium? (Z = 22)**

(a) 4s 3s 3p 3d (b) 3s 3p 3d 4s (c) 3s 3p 4s 3d (d) 3s 4s 3p 3d

Ans: (c)

58 **The number of d-electrons in Fe^{2+} (Z = 26) is not equal to the number of electrons in which one of the following?**

(a) d-electrons in Fe (Z = 26) (b) p-electrons in Ne (Z = 10)

(c) s-electrons in Mg (Z = 12) (d) p-electrons in Cl (Z = 17)

Ans: (d)

59 **The angular momentum of electrons in d-orbital is equal to**

(a) $2\sqrt{3}\,\hbar$ (b) $0\,\hbar$ (c) $\sqrt{6}\,\hbar$ (d) $\sqrt{2}\,\hbar$

Ans: (c) $\sqrt{6}\,\hbar$

Solution

$\sqrt{l(l+1)}\,\hbar$; for d-orbital, $l = 2$: $\sqrt{2(2+1)} = \sqrt{2 \times 3} = \sqrt{6}\,\hbar$

60 **What is the maximum number of electrons that can be associated with the following set of quantum numbers? n = 3, l = 1, m_l = 0**

(a) 1 (b) 2 (c) 3 (d) 4

Ans: (b) 2

61 **What is the maximum no. of orbitals that can be identified with the following quantum numbers? n = 3, l = 1, m = –1**

(a) 1 (b) 2 (c) 3 (d) 4

Ans: (a) 1

62 **The outer electronic configuration of Gd (Z = 64) is**

(a) $4f^5\,5d^4\,6s^1$ (b) $4f^7\,5d^1\,6s^2$ (c) $4f^3\,5d^5\,6s^2$ (d) $4f^4\,5d^5\,6s^1$

Ans: (b) $4f^7\,5d^1\,6s^2$

63 **Maximum no. of electrons in a subshell with l = 3 and n = 4 is**

(a) 14 (b) 16 (c) 10 (d) 12

Ans: (a)

64 **The correct set of four quantum numbers for valence electron of rubidium atom (Z = 37) is**

(a) 5, 1, 1, $+\frac{1}{2}$ (b) 6, 0, 0, $+\frac{1}{2}$ (c) 5, 0, 0, $+\frac{1}{2}$ (d) 5, 1, 0, $+\frac{1}{2}$

Ans: (c)

65 **The total no. of atomic orbitals in the fourth energy level of an atom is**

(a) 8 (b) 16 (c) 32 (d) 4

Ans: (b)

A
A
Verify

Items flagged for verification

Each tag in the text points to one row below: the page of the handwritten notes, the page of this book, and what should be checked. Most flags concern text cut off at the edge of the scan, faint writing, or values that look inconsistent in the original and were kept exactly as written.

Flag	Notes	Book	What to check
V1	p. 1	2	Tube length: the first digit is unclear in the handwriting; read as 50 cm (could be 60 cm).
V2	p. 4	12	The 'Uses' column is cut off at the right edge of the scan; words completed from the visible letters (scienti..., Cancer treatm..., Medica... Imag..., Sterlis..., Heat, remot..., Cooking radar, Broadcasting).
V3	p. 4	12	Red-end value is cut off at the scan edge (only '70...' visible); read as 700 nm.
V4	p. 4	12	The conversion '= 1368 × 10^{3} s^{-1}' is cut off at the right edge of the scan; completed from the next line of working.
V5	p. 4	13	Q1 (p. 4): the second part of the question (which part of the EM spectrum) is not answered in the original notes.
V6	p. 4	13	Q2 (p. 4): unit after 7.5 × 10^{14} is cut off at the scan edge; s^{-1} completed.
V7	p. 4	13	Q3 (p. 4): the bracket '(5800 × 10^{-10} m)' and the units at the ends of both answer lines are cut off at the scan edge; completed.
V8	p. 6	16	Einstein conclusion (ii): the relation symbol is overwritten in the handwriting; read as '=' since the line states K.E. = 0.
V9	p. 6	16	Q1 (p. 6): a stray '10^{-15}' is written beside this working, apparently rough work; it is not part of any step, so it is not reproduced. The numerator '× 3 × 10^{8} m s^{-1}' is cut off at the scan edge and was completed.
V10	p. 6	17	Q2 (p. 6): the end of the last line is cut off at the scan edge after '1.9878 ×'; '10^{-19} J' completed from 19.878 × 10^{-20}. Bracketed terms '7 × 10^{14} s^{-1}' are also partly cut off.
V11	p. 6	17	Spectroscope definition (p. 6): line ends 'different wave...' and 'dispersion of ...' are cut off at the scan edge; completed as 'wavelengths' and 'light'.
V12	p. 8	20	Merits ① and ② (p. 8): both lines are cut off at the right edge of the scan ('explain the stabili...', 'has been used to d...'). Completed as 'stability of atom' and 'derive Balmer's formula' (the derivation that follows is headed 'Balmer's formula'). Please verify.
V13	p. 8	21	Balmer derivation (p. 8): the right-hand ends of several lines are cut off at the scan edge (closing brackets / final terms); they were completed so each line is mathematically whole. Values shown (2.18 × 10^{-18} J, 0.329 × 10^{16}, 3.29 × 10^{15}, 1.09677 × 10^{7} m^{-1}, 1.09677 × 10^{5} cm^{-1}) are exactly as written.
V14	p. 8	21	Bohr merit ③ and limitations ①–⑤ (p. 8): several line endings are cut off at the scan edge ('atomic spe...', 'magnetic f...', 'presence of elec...', 'ability of atom...'). Completed words are shown with a dotted underline.
V15	p. 9	21	Q n = 5 → 2 (p. 9): the first line is written with '−2.18 × 10^{-18} J' while the next lines use '+2.18 × 10^{-18} J' (which gives the stated −4.58 × 10^{-19} J). Reproduced exactly as written; please check the sign in the first line.
V16	p. 10	24	Heisenberg conditions/explanation (p. 10): right-hand line endings are cut off at the scan edge ('microsc...', 'accura...', 'consider an el...', 'the moment...', 'energy is ...'). Completed words are dotted-underlined.
V17	p. 10	24	Heisenberg significance ①–④ (p. 10): line endings cut off at the scan edge ('simult...', 'Quantum mecha...', 'macroscopic ob...', 'in a fi...'); completed.

Flag	Notes	Book	What to check
V18	p. 10	24	Microscope question (p. 10): the ends of the question lines are cut off at the scan edge; wording completed ('...photon is employed... in an atom... What is the uncertainty... measurement of its velocity').
V19	p. 12	28	Spin quantum number (p. 12): line endings 'It is d...', 'gives the i...', 'spinning of ele...' and 'two possible ...' are cut off at the scan edge; completed.
V20	p. 12	28	Radial nodes formula (p. 12): cut off at the scan edge after '= n...'. Completed as n − l − 1 (the standard formula); please verify against the original.
V21	p. 12	28	Importance of radial wave function (i) (p. 12): line cut off after 'determining orbital...'; completed as 'size'.
V22	p. 13	29	Angular wave function symbol (p. 13): the handwritten letter before (θ, ϕ) could be read as 'Y' or 'Ψ'; shown as Y(θ, ϕ).
V23	p. 14	31	Aufbau (p. 14): the notes write '3d-orbital: n + l = 4 + 2 = 5'. For 3d, n = 3, so the working would normally read 3 + 2 = 5 (the result 5 is unchanged). Reproduced exactly as written; please verify.
V24	p. 14	31	Hund's rule paragraph (p. 14): several line endings are cut off at the scan edge ('when more th...', 'equal energies a...', 'will fi...', 'singly wi...', 'This is d...', 'electrons ca...', 'repel each', 'move f...'); completed.
V25	p. 14	31	Hund's rule 'Wrong' d^5 box diagram (p. 14): the arrows are crowded in the handwriting; drawn as ↑↓ ↑↓ ↑ _ _ (five electrons, paired before filling singly). Please verify the exact arrangement.
V26	p. 15	33	Exercise Q1 (p. 15): the top line of this page is cut off in the scan; only '...of electrons which will together weigh 1 gram' is visible. The opening words 'Calculate the number' were completed from context.
V27	p. 15	33	Q2(ii)(b) (p. 15): the notes end this line with '= 2.4088' and no power of ten ($24.088 \times 10^{20} = 2.4088 \times 10^{21}$). Reproduced as written.
V28	p. 15	34	Q2 (p. 15): the last part of (iii) — 'Will the answer change if the temperature and pressure are changed?' — is not answered in the original notes. Also, the step 'No. of protons in 34g of NH_3' is written as 34 g (the question says 34 mg); reproduced as written.
V29	p. 16	35	Q5 (p. 16): the frequency appears to be written as 5.17×10^{-14} s^{-1} (with a minus in the exponent); $c/\lambda = 3 \times 10^8 / 580 \times 10^{-9}$ works out to 5.17×10^{14} s^{-1}. Reproduced as read; please verify the exponent.
V30	p. 16	35	Q6 (p. 16): the right-hand ends of the question and of every answer line are cut off at the scan edge. Completed parts are dotted-underlined; values are taken from the visible working (19.878×10^{-19}, 1.9878×10..., 39.7..., 3.97...).
V31	p. 16	35	Q7 (p. 16): line endings cut off at the scan edge ('0.5 × 1...', '= 5 × 1...', '= 6...', '= 1.66 × ...'); completed from the working.
V32	p. 16	36	Q8 (p. 16): the last two lines are cut off at the scan edge ('= 0.201...', '= 2.012 × 1...'); completed from $1 \div (4.969 \times 10^{-17})$.
V33	p. 17	36	Q9 (p. 17): the left edge of this page is cut off in the scan, so the question number and the first letters of several lines ('[st]rikes', '[wor]k function', '[2.]13 eV') are missing; completed from context (2.13 eV is confirmed by the working).
V34	p. 17	37	Q13 (p. 17): reproduced as written — 109678 × 12/64 is written as 20564.28 cm^{-1} (direct multiplication gives ≈ 20564.6), and '486×10^{-7} cm^{-1}' carries the unit cm^{-1} where the quantity is a length. Final answer 486 nm unaffected. Please verify.
V35	p. 18	38	Q14 (p. 18): the comparison ratio is written as IE_1/IE_2 although it compares IE_1 with IE_5 (8.72×10^{-20} J). Reproduced as written.
V36	p. 18	38	Q15 (p. 18): the end of the question is cut off at the scan edge; 'in n = 6' is completed from the working, which uses n = 6.
V37	p. 18	38	Q16 (p. 18): several line endings are cut off at the scan edge ('−2.17×...', 'J a...', '−0.0868 × 1...', '= 13.225...'); completed from the working. The unit of the final radius (shown as Å) is not visible.

Flag	Notes	Book	What to check
V38	p. 18	39	Q17 (p. 18): the line after 15232.9 cm^{-1} is written '= 1.52 × 10^{-4} cm^{-1}' (15232.9 = 1.52 × 10^{4}). The next line, 1.52 × 10^{6} m^{-1}, is consistent with 10^{4} cm^{-1}. Reproduced as written; please verify the sign of the exponent.
V39	p. 19	39	Q18 and Q19 (p. 19): the left edge of this page is cut off in the scan, so the question numbers and the first letters of several lines ('[to] shift', '[at]om', '[fif]th orbit', '[ret]urns', '[sta]te', '[E_1] =', '[E_5] =', '[ΔE] =') are missing; completed from context. A minus sign before 2.18 × 10^{-11} erg is not visible.
V40	p. 20	42	Q28 (p. 20): the top of the right column is cut off in the scan; the question is completed as 'An atomic orbital has n = 3. What are possible values of l [and m_l]?' (n = 3 is visible in the working). The ends of the m_l lines are also cut off.
V41	p. 21	43	Q30 (iv) (p. 21): the top line of this page is cut off in the scan, so the value of n in set (iv) is not visible. The answer reads 'Not possible because for l = 0, m_l cannot be +1. It must [be] zero.'
V42	p. 21	43	Q30 (p. 20): the ends of answers (i) and (iii) are cut off at the scan edge ('n cannot [be zero]', 'l [cannot] be equal to 1'); completed.
V43	p. 21	44	Q33 (p. 21): written as 'For He^{+}-ion, Z = 1', but the next line uses $Z^2 = 2^2$ (He^{+} has Z = 2). Reproduced as written; please verify.
V44	p. 22	45	Q39 (p. 22): the first line of the question is cut off at the top of the scan; only '...charge on oil droplet was [...] to be −1.282 × 10^{-18} C' is visible. Opening words completed from context.
V45	p. 22	45	Q40 (p. 22): line endings of the question and answer are cut off at the scan edge ('heavy [...]', 'by the α-[particles]', 'like Alumin[ium]', 'what diff[erence]', 'deflect[ion]', 'heavy meta[ls]', 'positive cha[rge]'); completed.
V46	p. 22	46	Q41 (p. 22): the second 'correct' symbol appears to be written with only the lower number (35Br), while the question quotes ^{79}Br. Reproduced as read; please verify.
V47	p. 24	48	Q50 and Q51 (p. 24): the right column of this page is heavily cut off at the scan edge (ends of the question lines, units, and the last digits of several results such as '5.0934 × 1...', '5.08...', '3.43...', '1.9 × 1.602 × 1...', '3 × 10^{8}/4.59...'). Completed from the visible working; please verify against the original.
V48	p. 25	49	Q51 (p. 24–25): the bracket in '½mv^2 = ... hc[1/λ − 1/λ_0]' is cut off at the scan edge (shown as [...]); the next line uses 1/500 − 1/654. The top of p. 25 is also slightly cut.
V49	p. 26	51	Q56 (p. 26): reproduced as written — the ratio n_1/n_2 = 2.5 is followed by 'if n_1 = 2, n_2 = 5'; the line endings 'This transi...', 'from orbit...', 'belongs to Bal...' are cut off at the scan edge and completed.
V50	p. 27	52	Q59–Q61 (p. 27): the left edge of this page is cut off in the scan, so the first letters of several lines are missing; completed from context.
V51	p. 27	52	Q61 (p. 27): the 'actual momentum' step is written with 0.5 × 10^{-9} m in the denominator, while the question gives 0.05 nm; the stated result 1.055 × 10^{-24} kg m s^{-1} corresponds to 0.05 × 10^{-9} m. Reproduced exactly as written; please verify.
V52	p. 29	56	MCQ Set 1 (p. 29): the left edge of this page is cut off in the scan; the first characters of some lines (option labels, 'Ans') are missing. Q4 has no answer written in the original.
V53	p. 30	57	Set 2 Q3 (p. 30): option (iv) is cut off at the column edge; read as O_2 (a charge sign may be missing).
V54	p. 30	58	Set 2 Q8 (p. 30): the last pairing in options (ii) and (iv) is cut off at the scan edge (shown as 'D-...'). Pair (C) is written as 'H, H^{+}'.
V55	p. 30	59	Set 2 Q10 (p. 30): the last species in the assertion is cut off at the scan edge (only '$^{5}_{3}$...' visible); shown as $^{5}_{3}Li$.
V56	p. 31	60	MCQ Set 3 (p. 31): the left edge of this page is cut off in the scan, so the first characters of many lines (question words, option labels) are missing; completed from context. No answers are filled in on this page.
V57	p. 32	62	NEET PYQ 6 (p. 32): the end of the question line is cut off at the scan edge ('isoelectronic w...'), completed as 'with CO'; option (d) is partly cut and read as N_2^{-}.

Flag	Notes	Book	What to check
V58	p. 32	62	NEET PYQ 8 (p. 32): the right-hand ends of the solution lines are cut off at the scan edge ('= R_H[...', '= 3...'); completed from the working.
V59	p. 32	62	NEET PYQ 9 (p. 32): option (d) is cut off at the scan edge (only '21...' visible).
V60	p. 33	63	NEET PYQs 10–13 (p. 33): the left edge of this page is cut off in the scan, so question numbers and the first letters of several lines are missing; completed from context.
V61	p. 34	64	NEET PYQs 17–21 (p. 34): this page is written in faint ink and the scan is low-contrast; all values were read carefully and cross-checked against the working, but the exponents in the options of Q19 (10^{-16} vs 10^{-18}) and the option values of Q18 are faint. Please verify.
V62	p. 34	66	NEET PYQ 21 (p. 34): the last value is cut off at the scan edge after '1.58...'; completed as 1.587 Å.
V63	p. 35	66	NEET PYQs 22–24 (p. 35): the top and left edge of this page are cut off in the scan; question numbers and first letters of several lines are missing and were completed from context.
V64	p. 35	67	NEET PYQ 25 (p. 35): the end of the second ΔE line is cut off at the column edge (shown as '[1/16 – ...]').
V65	p. 36	68	NEET PYQ 27 (p. 36): the option letter written before 'Sommerfeld' in the answer is faint; the answer is shown as (d) Sommerfeld, matching the option list.
V66	p. 36	68	NEET PYQ 29 (p. 36): the answer is written as '(c) 0.1936×10^{-10} ergs', but 0.1936×10^{-10} ergs is option (d) in the list. Reproduced as written; please verify.
V67	p. 36	68	NEET PYQ 30 (p. 36): option (d) is cut off at the scan edge (only '0.2...' visible).
V68	p. 37	69	NEET PYQs 35–37 (p. 37): the left edge of this page is cut off in the scan; question numbers and first letters of lines are missing and were completed from context.
V69	p. 37	69	NEET PYQ 36 (p. 37): option (d) is cut off at the column edge; only its beginning (√h...) is visible.
V70	p. 37	70	NEET PYQ 39 (p. 37): the last line of the solution appears to read 5.27×10^{-28} m (faint), whereas the chosen option and the calculation give 5.27×10^{-30} m. Reproduced as read; please verify.
V71	p. 38	71	NEET PYQ 42 (p. 38): the solution writes h as 6.626×10^{-34} (and 'J s' in the numerator), while the question gives h = 6.626×10^{-27} ergs; the result 1.93 cm corresponds to 6.626×10^{-27} erg s. Reproduced exactly as written; please verify.
V72	p. 39	73	NEET PYQ 46 (p. 39): the top of this page and its left edge are cut off in the scan, so the opening words of the question and the first characters of the options are missing. The visible text and the answer (b) with its working are reproduced.

B
Checklist

Page-by-page verification checklist

All 40 pages of the handwritten notes were read in full (dense calculations were re-read at high magnification) and transcribed in their original order. The table shows what each original page contains, how it was laid out here, how many verification flags it carries and where it appears in this book.

Notes	Content	Laid out as	Flags	Book pages	Status
p. 1	Atom; sub-atomic particles; discharge tube experiment (set-up, 3 observations, fluorescence); 9 properties of cathode rays; conclusion; charge and mass of electron; discovery of proton; 6 properties of anode rays	6 cards, 1 diagrams	1	4–6	✓ Done
p. 2	Discovery of neutrons; Thomson's model and limitation; Rutherford model (1911); α-scattering experiment with 2 diagrams; observations; conclusions; nucleus; 9 features; planetary analogy	6 cards, 3 diagrams	0	6–8	✓ Done
p. 3	Failure of Rutherford model (4 points); atomic number; mass number; Q1–Q3; isotopes, isobars, isotones; EM wave theory; 5 characteristics of EM waves	3 numericals, 5 cards, 1 tables, 1 diagrams	0	8–11	✓ Done
p. 4	5 characteristics of waves; λ–ν–$\bar{\nu}$ relations; EM spectrum and its order; table of 8 radiations; visible spectrum; Q1–Q3	3 numericals, 9 cards, 1 tables, 2 diagrams	6	11–13	✓ Done
p. 5	Planck's quantum theory (4 features); Q1–Q2; black body radiation, temperature–colour table; photoelectric effect, observations 1–3	2 numericals, 4 cards, 1 tables, 1 diagrams	0	13–15	✓ Done
p. 6	Observation 4; Einstein's explanation and equation; conclusions; Q1 (sodium, 517 nm); Q2 (K.E.); atomic spectra; spectroscope; emission spectra	2 numericals, 6 cards	4	15–17	✓ Done
p. 7	Continuous and line spectra; absorption spectra; line spectrum of hydrogen with series diagram; Rydberg formula; Q (n = 4 → 2)	1 numericals, 4 cards, 2 diagrams	0	17–19	✓ Done
p. 8	Bohr's model: 6 postulates; merits; Balmer's formula derivation; 5 limitations	8 cards, 1 diagrams	3	19–21	✓ Done
p. 9	Q (n = 5 → 2); Q (He^+ first orbit); particle vs wave table; de Broglie relation, derivation and significance	2 numericals, 2 cards, 1 tables	1	21–23	✓ Done
p. 10	Q (ball wavelength); Q (golf ball Δx); Heisenberg principle: statement, formula, conditions, explanation, significance; Q (microscope Δv); quantum mechanical model	3 numericals, 4 cards	3	23–25	✓ Done
p. 11	Main postulates 1–6 with Schrödinger equation; atomic orbital and cloud diagram; orbit vs orbital table; quantum numbers; principal quantum number	4 cards, 1 tables, 1 diagrams	0	25–27	✓ Done
p. 12	Azimuthal (l values, subshell table), magnetic and spin quantum numbers; radial wave function, graph, radial nodes, importance	5 cards, 2 tables, 1 diagrams	3	27–29	✓ Done
p. 13	Angular wave function; shapes of s, p and d orbitals with 3 diagrams	1 cards, 3 diagrams	1	29–30	✓ Done
p. 14	Aufbau principle with filling-order diagram; Pauli's principle table; Hund's rule box diagrams; stability of filled and half-filled orbitals; Cr and Cu	8 cards, 1 tables, 1 diagrams	3	30–33	✓ Done

Notes	Content	Laid out as	Flags	Book pages	Status
p. 15	Exercise Q1–Q2	2 numericals	3	33–34	✓ Done
p. 16	Exercise Q3–Q8	6 numericals, 1 tables	4	34–36	✓ Done
p. 17	Exercise Q9–Q13	5 numericals	2	36–38	✓ Done
p. 18	Exercise Q14–Q17	4 numericals	4	38–39	✓ Done
p. 19	Exercise Q18–Q23 (i)	6 numericals, 1 tables	1	39–41	✓ Done
p. 20	Exercise Q23 (ii)–(iii), Q24–Q30 (iii)	8 numericals, 4 tables	1	41–43	✓ Done
p. 21	Exercise Q30 (iv)–(vi), Q31–Q35	5 numericals	3	43–44	✓ Done
p. 22	Exercise Q36–Q41, start of Q42	7 numericals	3	44–46	✓ Done
p. 23	Exercise Q42 (contd.)–Q47	5 numericals	0	46–47	✓ Done
p. 24	Exercise Q48–Q51 (part)	4 numericals	1	47–49	✓ Done
p. 25	Exercise Q51 (contd.)–Q54	3 numericals, 1 tables	1	49–50	✓ Done
p. 26	Exercise Q55–Q58	4 numericals	1	50–52	✓ Done
p. 27	Exercise Q59–Q65	7 numericals, 1 tables	2	52–53	✓ Done
p. 28	Exercise Q66–Q67 (rest of page blank)	2 numericals, 1 tables	0	53–55	✓ Done
p. 29	Practice MCQs Set 1, Q1–Q10	10 MCQs	1	55–57	✓ Done
p. 30	Practice MCQs Set 2, Q1–Q10	10 MCQs, 1 tables	3	57–59	✓ Done
p. 31	Practice MCQs Set 3, Q1–Q10	10 MCQs	1	59–61	✓ Done
p. 32	NEET PYQs 1–9	9 MCQs	3	61–63	✓ Done
p. 33	NEET PYQs 10–16	7 MCQs	1	63–64	✓ Done
p. 34	NEET PYQs 17–21	5 MCQs	2	64–66	✓ Done
p. 35	NEET PYQs 22–26	5 MCQs	2	66–68	✓ Done

Notes	Content	Laid out as	Flags	Book pages	Status
p. 36	NEET PYQs 27–34	8 MCQs	3	68–69	✓ Done
p. 37	NEET PYQs 35–40	6 MCQs	3	69–71	✓ Done
p. 38	NEET PYQs 41–45	5 MCQs, 1 tables	1	71–73	✓ Done
p. 39	NEET PYQs 46–53	8 MCQs	1	73–74	✓ Done
p. 40	NEET PYQs 54–65	12 MCQs	0	74–84	✓ Done

C
C
Figures

Diagrams

Redrawn from the original notes

Diagram	Notes page
Rutherford's α-scattering set-up	p. 2
Paths of α-particles near the nucleus	p. 2
Spectral series of hydrogen	p. 7
Cloud representation of electron	p. 11
Radial wave function of 1s orbital	p. 12
1s, 2s, 3s orbitals with nodes	p. 13
Shapes of p-orbitals	p. 13
Shapes of the five d-orbitals	p. 13
Sequence of filling of orbitals	p. 14

Illustrations added in this edition

These do not appear in the handwritten notes. They only picture what the accompanying text already says and add no new facts; they can be removed without losing any original content.

Illustration	Placed with notes page
Discharge tube	p. 1
Thomson's plum-pudding model	p. 2
Electron spiralling into nucleus	p. 3
Parts of a wave	p. 4
EM spectrum bar	p. 4
Photoelectric effect	p. 5
Continuous, line and absorption spectra	p. 7
Bohr's orbits	p. 8

End of book